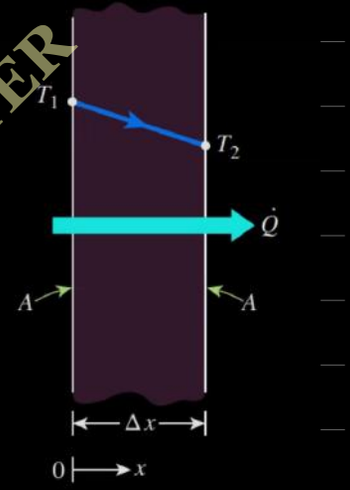
Heat Transfer: In accordance with second law of thermodynamics, transfer of energy in form of heat always occurs from higher temperature to lower temperature. Heat transfer modes can be classified in to conduction, convection and radiation.

Conduction: In this mode of tranfer of heat, energy is transfered to nearby particles either through lattice vibrations or energy transport by free electrons (metals). In this mode there is no bulk transport of material. The rate of heat conduction depends on geometry of the medium, its material, its thickness as well as the temperature difference across the medium.



For instance consider steady state heat conduction through a wall of thickness Δx, Area A and temperature difference $\Delta T = T_2 - T_1$. Experiments show that rate of heat transfer is doubled on doubling the temperature across the wall and halved on doubling the thickness. Thus rate of heat transfer by conduction is proportional to the temperature difference, inversely proportional to thickness of the layer and directly proportional to the Area. Mathematically:

$\dot{Q} = kA\left(\frac{T_1 - T_2}{\Delta x}\right)$ or $\dot{Q}_{cond} = -kA\left(\frac{\Delta T}{\Delta x}\right)$. Here the constant k depends on the material and is known as thermal conductivity of the material. In the limiting case when $\Delta x \to 0$, the equation becomes:

$$\dot{Q}_{cond} = -kA\frac{dT}{dx}$$

This is known as Fourier's law of Heat Conduction (One dimensional heat conduction equation).

→ Thermal Conductivity of crystalline substances is better than amorphous substances.

→ Thermal conductivity of metals is much higher than non metals due to presence of free electrons (They can quickly carry energy within the metall lattice due to being free to move).

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Example (Cengel):

EXAMPLE 1–5 The Cost of Heat Loss Through a Roof

The roof of an electrically heated home is 6 m long, 8 m wide, and 0.25 m thick, and it is made of a flat layer of concrete whose thermal conductivity is $k = 0.8$ W/m·K (Fig. 1–27). The temperatures of the inner and the outer surfaces of the roof one night are measured to be 15°C and 4°C, respectively, for a period of 10 h. Determine (*a*) the rate of heat loss through the roof that night and (*b*) the cost of that heat loss to the homeowner if the cost of electricity is $0.08/kWh.

Solution: $\dot{Q} = -kA\left(\frac{\Delta T}{\Delta n}\right)$

$= 0.8 \times 6 \times 8 \times \frac{(15 - 4)}{0.25} = 1.69\ kW$

$\therefore$ In 10 Hrs $E = \dot{Q} \times t = 1.69 \times 10$ kwhr

$\Rightarrow E = 16.9$ kWh.

$\Rightarrow$ Cost $= 0.08 \times 16.9 = \$1.35$. Ans.

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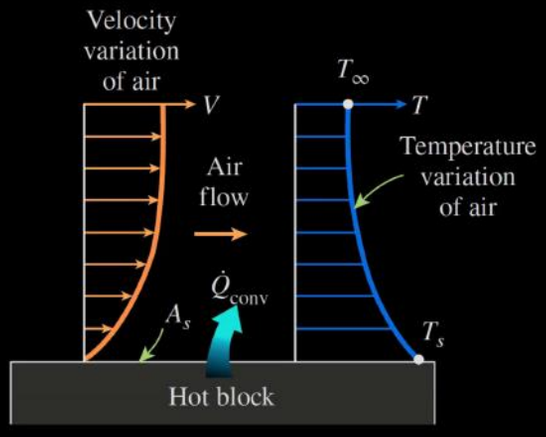
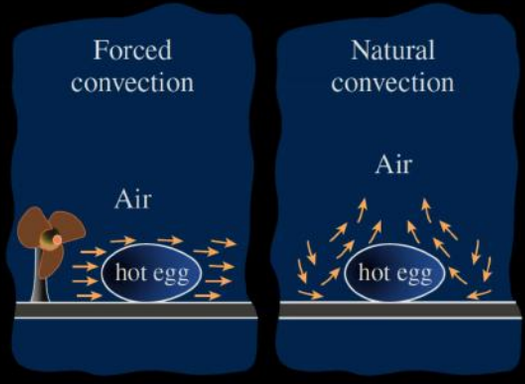
Convection: This mode of heat transfer occurs due to bulk motion of fluid particles carrying energy from one place to another. Convection can be further divided in to two modes:

Forced Convection: In this mode of convection heat transfer, fluid is forced to flow using external means such as a fan, pump or wind.

Natural (or free) Convection: Here fluid motion is caused by buoyancy forces that are induced by density differences due to variation of temperature within the fluid.





Newton's Law Of Cooling: In general, physics of convection is

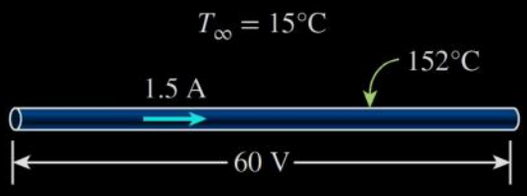
quite complicated however rate of convective heat transfer is also observed to be proportional to the temperature difference from surroundings and area of the cooling surface. Mathematically:

$$\dot{Q}_{conv} = h A_s (T_s - T_\infty)$$

Here h is called convection heat transfer coefficient which depends on many variables influencing convection like, surface geometry, nature of fluid flow, properties of fluid and bulk fluid velocity. h is usually found experimentally due to it complex dependence on so many variables.

EXAMPLE (CENGEL):

A 2-m-long, 0.3-cm-diameter electrical wire extends across a room at 15°C, as shown in Fig. 1–37. Heat is generated in the wire as a result of resistance heating, and the surface temperature of the wire is measured to be 152°C in steady operation. Also, the voltage drop and electric current through the wire are measured to be 60 V and 1.5 A, respectively. Disregarding any heat transfer by radiation, determine the convection heat transfer coefficient for heat transfer between the outer surface of the wire and the air in the room.



Solution: $\dot{Q}_{conv} = h A_s (T_s - T_\infty)$

$$\Rightarrow \quad VI = h \times 2\pi r l (T_s - T_\infty)$$

$$\Rightarrow h = VI / [2\pi r l (T_s - T_\infty)] = 60 \times 1.5 / (2\pi \times \tfrac{3}{2} \times 10^{-3} \times 2 \times 137)$$

$$\Rightarrow h = 34.9 \ W/m^2K.$$

→ This assumes all heat transfer through convection (Ignoring any heat transfer by radiation).

Radiation: When energy is transferred from a hot surface to cold surface through electromagnetic waves we call it radiation. Maximum rate at which radiation can be emitted from a surface is given by Stefan-Boltzmann law as:

$$\dot{Q}_{emit,max} = \sigma A_s T_s^4$$

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Here $\sigma = 5.67 \times 10^{-8}\ W/m^2K^4$, A_s is the area of radiating surface, T_s is the temperature of radiating surface. The idealized surface which emits radiation at this rate is called a **BLACK BODY**. Radiation emitted by all real surfaces is less than the radiation emitted by a blackbody at the same temperature and is expressed as:

$$\dot{Q}_{emit} = \epsilon \sigma A_s T_s^4$$

Here ϵ is called emissivity of the surface. For any surface, $0 \leq \epsilon \leq 1$ and it is a measure of how closely the surface approximates a black body.

ABSORPTIVITY: The fraction of incident energy on any surface which gets absorbed is called absorptivity α of the surface.

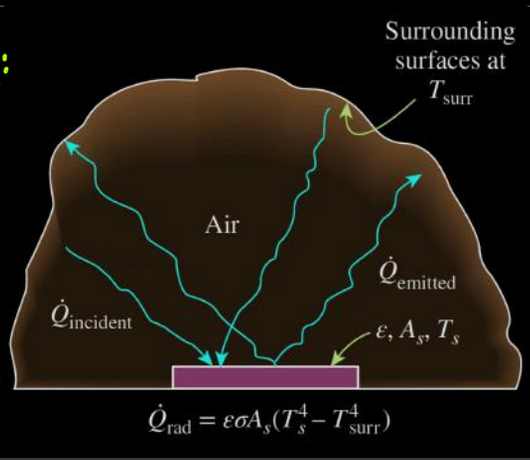
Mathematically: $\dot{Q}_{absorbed} = \alpha \dot{Q}_{incident}$.

Like ϵ absorbtivity α of any surface is also lies between 0 & 1 i.e $0 \leq \alpha \leq 1$.

KIRCHOFF'S LAW OF RADIATION: The emissivity and absorptivity of any surface at a given temperature and wavelength are equal

Mathematically, $\epsilon = \alpha$

NET RADIATIVE HEAT TRANSFER FOR ENCLOSED SURFACE:



When a surface of emissivity ϵ, Surface Area A_s and thermodynamic temperature T_s is completely enclosed by a much larger (or black) surface at a thermodynamic temperature T_{surr} separated by a gas (such as air) that does not intervene with radiation, the net rate of radiation heat transfer between these two surfaces is given as:

$$\dot{Q}_{rad} = \epsilon \sigma A_s (T_s^4 - T_{surr}^4) \quad (1)$$

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Note that if surrounding surface is much larger than the radiating surface, the emissivity of surrounding surface does not matter.

→ Equation ① holds only if the radiating surface is flat or convex, a concave also radiates to itself and we shall deal with it later on.

COMBINED HEAT TRANSFER: In general, heat transfer will be occuring combinedly through all modes and we can write :

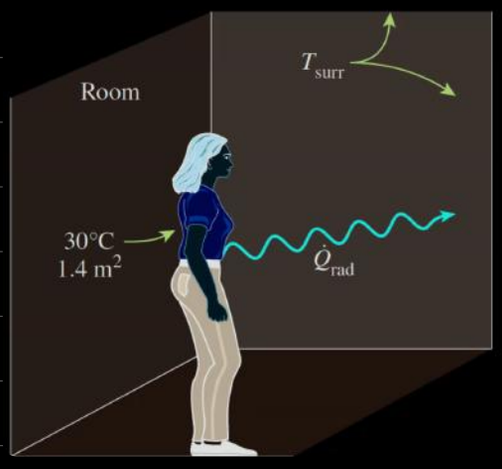
$$\dot{Q}_{total} = \dot{Q}_{conv} + \dot{Q}_{rad} = h_{conv} A_s (T_s - T_{surr}) + \epsilon \sigma A_s (T_s^4 - T_{surr}^4)$$

$$\Rightarrow \dot{Q}_{total} = h_{combine} A_s (T_s - T_{surr})$$

where $h_{combined} = h_{conv} + \epsilon \sigma (T_s + T_{surr})(T_s^2 + T_{surr}^2)$.

For most practical situations, radiation effects are significant relative to conduction or natural convection but negligible relative to forced convection. Radiation effects are generally disregarded in forced convection especially if surfaces have low emissivities and low to moderate temperatures.

Example: (Cengel): A person is standing in a room maintained at 22°C at all times. Inner surfaces of walls ceiling and floor of the room are 10°C in winter and 25°C in summers. Determine rate of radiation heat transfer between the person and surrounding if exposed surface area is $1.4\ m^2$ and surface temperature of person is 30°C. (Ignore convection effects and consider $\epsilon = 0.95$ for human body).



Solution: $\dot{Q}_{rad,winter} = \epsilon \sigma A_s (T_s^4 - T_{surr}^4)$

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$= 0.95 \times 5.67 \times 10^{-8} \times 1.4 \,(303^4 - 283^4)\ W = 152\ W$ Ans.

Similarly for summers:

$\dot{Q}_{rad,\,summer} = 0.95 \times 5.67 \times 10^{-8} \times 1.4\,(303^4 - 298^4) = 40.9\ W$ Ans.

(Note that emitted power in both cases is same, however in summers, the body is also receiving more power from surroundings)

→ This also explains why we feel more hungry in winters.

→ Always remember to convert temperatures to Kelvin while dealing with radiation heat transfers).

→ Net heat radiated is 4 times in winters as compared to summers this explains the chill felt in winters.

Example: In the situation of previous example suppose a breeze is blowing (say due to A.C. or an open window) find the convection heat transfer if Convection heat transfer coefficient is $h_{conv} = 6\ W/m^2$.

Sol: $\dot{Q}_{conv} = h A_s (T_s - T_{sur}) = 6 \times 1.4 \times (30°C - 22°C)\ W$

$\Rightarrow \dot{Q}_{conv} = 67.2\ W.$, Net $\dot{Q} = \dot{Q}_{conv} + \dot{Q}_{rad}$.

→ Heat transfer would be much higher if the person were not dressed since the exposed surface Area would be higher, thus an important function of clothes is to serve as a barrier against heat transfer.

→ There will also be some heat transfer through conduction to the floor through the feet which is usually very small and hence neglected.

→ Besides these modes heat transfer, there is also heat transfer from skin by perspiration, which is a dominant mode in hot environments.

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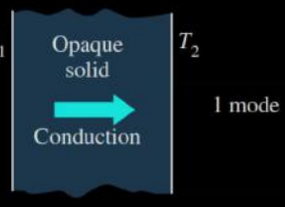
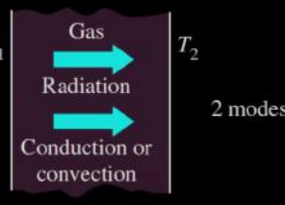
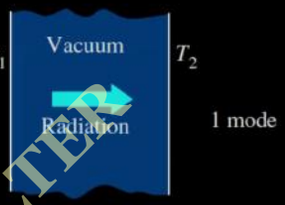
Considerations In Combined Heat transfer Modes:



→ If surfaces are separated by an opaque solid, there is no direct radiation from one surface to the other and heat transfer occurs through conduction only.



→ If surfaces are separated by a gas, the transfer occurs through Radiation as well as convection.

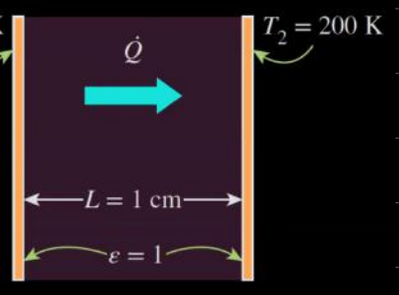


→ If surfaces are separated by vacuum, there is only radiative heat transfer.

Example (Cengel):

EXAMPLE 1–11 Heat Transfer Between Two Isothermal Plates

Consider steady heat transfer between two large parallel plates at constant temperatures of $T_1 = 300$ K and $T_2 = 200$ K that are $L = 1$ cm apart, as shown in Fig. 1–44. Assuming the surfaces to be black (emissivity $\varepsilon = 1$), determine the rate of heat transfer between the plates per unit surface area, assuming the gap between the plates is (*a*) filled with atmospheric air, (*b*) evacuated, (*c*) filled with urethane insulation, and (*d*) filled with superinsulation that has an apparent thermal conductivity of 0.00002 W/m·K.



(Assume steady state, $\epsilon = 1$ and no natural convection currents).

Sol: (a) $\dot{Q}_{cond} = \dfrac{kA(T_1 - T_2)}{L} = 0.0219 \times 1 \times \dfrac{(300 - 200)}{0.01\ m} = 219\ W.$

($K_{air} = 0.0219$ W/m K).

$\dot{Q}_{rad} = \epsilon \sigma A (T_1^4 - T_2^4) = 1 \times 5.67 \times 10^{-8} \times (300^4 - 200^4) = 369\ W$

$\Rightarrow \dot{Q} = \dot{Q}_{rad} + \dot{Q}_{cond} = 219 + 369 = 588$ W ans.

(b) When there is no air there is only radiation $\Rightarrow \dot{Q} = 369$ W Ans.

(c) For Urethane, $k = 0.026$ W/m K. Here there is only conduction.

$\therefore \dot{Q}_{cond} = \dfrac{0.026 \times (300 - 200) \times 1}{0.01} = 260\ W.$

→ Note that despite higher conductivity net heat transfer through Urethane is less than that of air due to cutting off the radiation losses.

(d) $\dot{Q}_{cond} = 0.00002 \times 1 \times \dfrac{(300 - 200)}{0.01} = 0.2\ W.$

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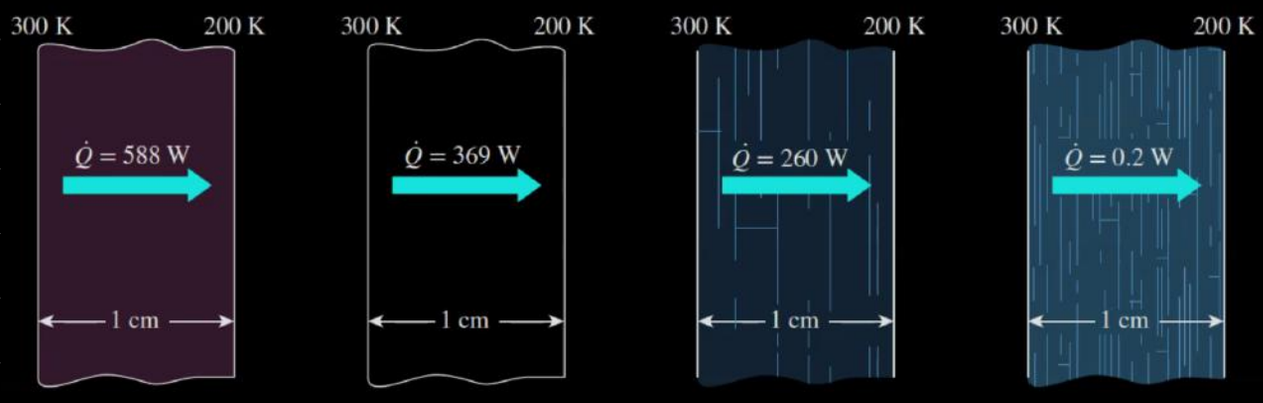
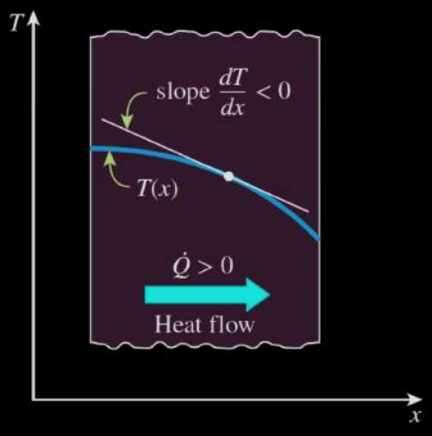


→ The above illustration highlights why choice of insulation can be very important in air conditioning, Refrigeration or ovens. It can make a big difference to operations cost.

CONDUCTION (DETAILED):

HEAT CONDUCTION IN THREE DIMENSIONS

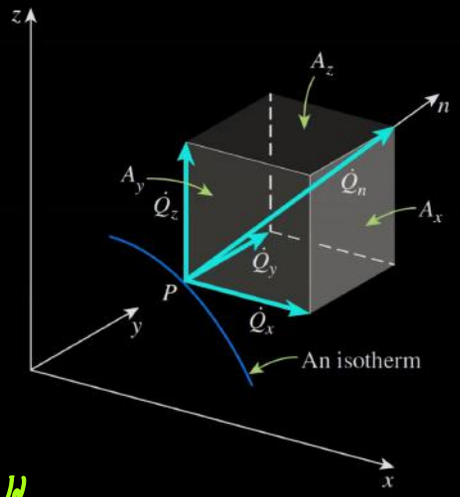


Recall Fourier's Law of Heat Conduction which related heat transfer in one dimension to the temperature gradient in the direction of heat flow.

$$\dot{Q} = -kA\left(\frac{dT}{dx}\right)$$

The same equation can be generalised when there is a three dimensional temperature gradient.

Consider a small material cube element having Areas A_x, A_y & A_z as shown. Let $\dot{Q}_x$, $\dot{Q}_y$ and $\dot{Q}_z$ be the heat transfer rates in x, y and z directions then:



$$\dot{Q}_x = -kA_x\frac{\partial T}{\partial x}, \quad \dot{Q}_y = -kA_y\frac{\partial T}{\partial y}, \quad \dot{Q}_z = -kA_z\frac{\partial T}{\partial z}$$

Here we assumed isotropic material but we could also have different k along different directions.

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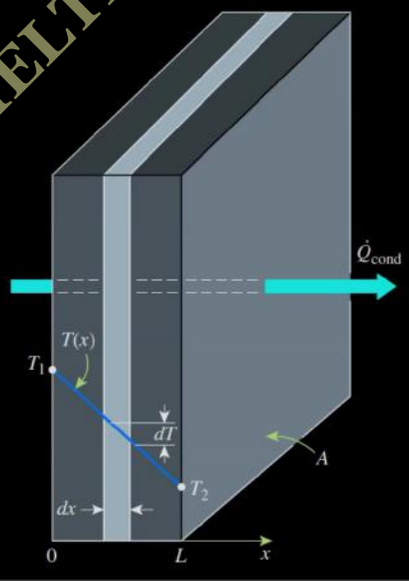
Steady State One Dimensional Conduction:

When temperatures don't change with time we call the problem as steady state conduction problem. In steady state problems, Heat outflow must be equal to the heat input if there is no generation and if there is generation then output must be the sum of input and generation. Usually for most systems it will take some time to attain steady state as initially some of heat flows will be causing internal energy changes to the system.

Analogy Between Electrical & Thermal Concepts:

Recall one dimensional heat transfer equation:

$\dot{Q} = -kA\frac{dT}{dx}$, If we have a wall



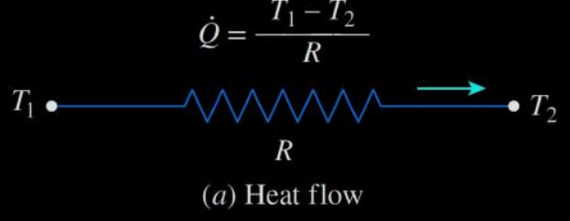
then under steady state conduction, this equation transforms in to:

$$\dot{Q} = kA\frac{(T_1 - T_2)}{L}$$

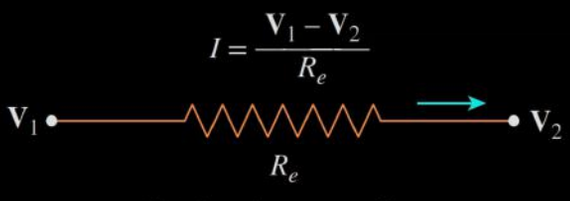
We could also consider this as a ratio of Thermal "potential difference" and a thermal resistance i.e. $\dot{Q} = \frac{T_1 - T_2}{R}$ where

R is the thermal resistance given as:

$$\boxed{R = L/kA}$$



(a) Heat flow



(b) Electric current flow

Thus we have following analogies:

Thermal	Electrical
Temperature (T)	Potential (V)
Heat transfer Rate ($\dot{Q}$)	Current (I)
Heat Q	Charge (q).
Thermal Resistance (R)	Electrical Resistance (R_e)
$R = L/kA$	$R = \rho L/A = L/\sigma A$
Thermal Conductivity (k)	Electrical conductivity σ.

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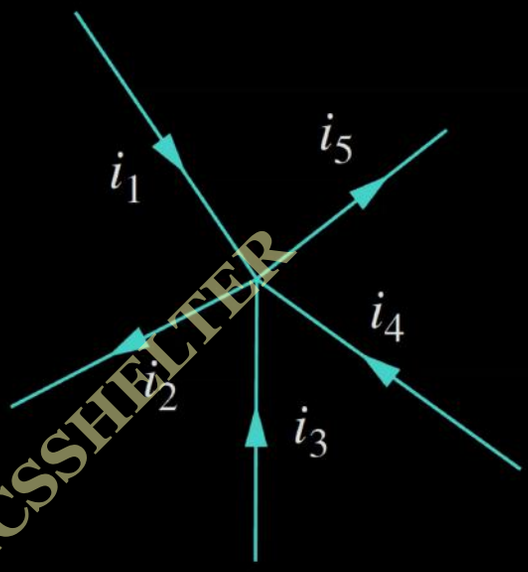
Having stated this analogy, we can apply the circuit theory of electrical circuits to thermal circuits (Concepts like Kirchoff's Voltage law, Junction law, Node Voltage Method, Wheatstone Bridge, series and parallel etc.).

KIRCHOFF'S LAWS FOR STEADY STATE HEAT FLOW:

KIRCHOFF'S JUNCTION LAW: In steady state net heat entering a junction is equal to net heat leaving a junction. For example in the shown adjacent figure:

$$i_1 + i_3 + i_4 = i_2 + i_5.$$



(Alternately we can say that algebraic sum of heats entering a junction is zero, here entering currents are treated as having positive signed values).

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EXAMPLE (JUNCTION LAW)

Example 22.10 *Three rods made of the same material and having the same cross-section have been joined as shown in the figure. Each rod is of the same length. The left and right ends are kept at 0°C and 90°C, respectively. The temperature of junction of the three rods will be* (JEE 2001)

(a) 45° C (b) 60° C
(c) 30° C (d) 20° C

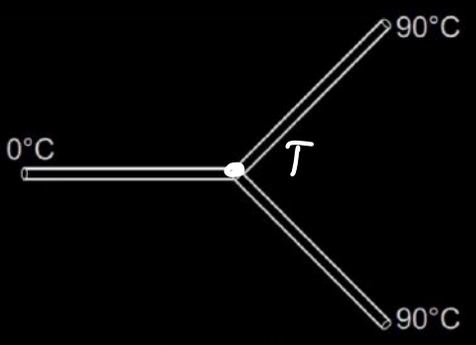
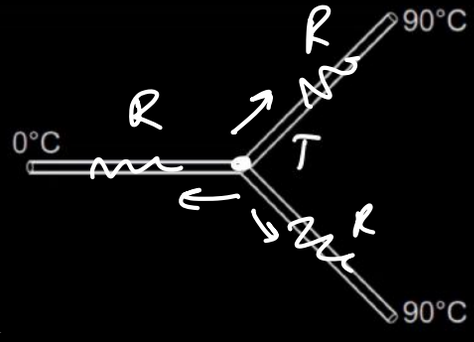


Fig. 22.14

Solution: Let the resistance of each rod be R

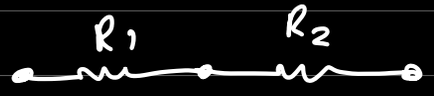
$$\Rightarrow \frac{T-0}{R} + \frac{T-90}{R} + \frac{T-90}{R} = 0$$

$$\Rightarrow T = 60°C \text{ Ans.}$$



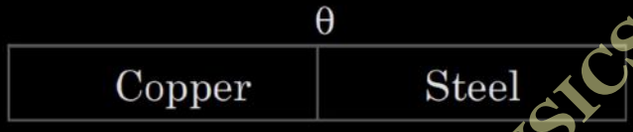
Series Combination Of Resistors: Two or more resistors are said to be in series if they carry same heat flow by virtue of configuration:

e.g. For adjacent figure, R_1 & R_2 are in series and can be replaced by an effective resistance given by $\boxed{R_{eff} = R_1 + R_2}$



Example (Resistors In Series)

Example 22.11 *Figure shows a copper rod joined to a steel rod. The rods have equal length and equal cross- sectional area. The free end of the copper rod is kept at 0° C and that of the steel rod is kept at 100° C. Find the temperature θ at the junction of the rods. Conductivity of copper = 390 W/m-° C and that of steel = 46 W/m-° C.*



Solution: Since the resistors are in series, they carry same heat flow.

$$\Rightarrow \frac{100-\theta}{(L/K_{St}A)} = \frac{\theta-0}{(L/K_{Cu}A)} \Rightarrow 46(100-\theta) = 390(\theta)$$

$\Rightarrow \theta = 10.6^\circ C$ Ans.

(Alternatively we could use the fact that in series temperature drops are in direct ratios of resistances. Thus temperature drop across copper:

$$\frac{R_{Cu}}{R_{Cu}+R_{St}} \times (100) = \frac{1/390}{(1/390)+(1/46)} \times 100 = 10.55^\circ C.)$$

Resistors In Parallel: Two or more resistors are said to be connected in parallel if they are connected across same temperature difference by virtue of configuration.

For example in the adjacent figure R_1 & R_2 are connected in parallel.

Here: $\boxed{\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2}}$

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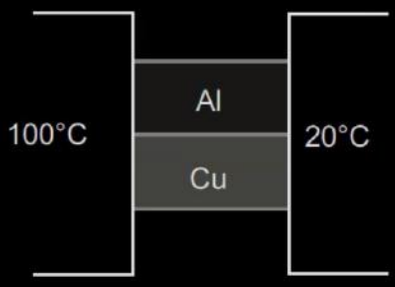
Example (Resistors In Parallel) :

Example 12 *Two metal cubes with 3 cm-edges of copper and aluminium are arranged as shown in figure. Find*
(a) the total thermal current from one reservoir to the other
(b) the ratio of the thermal current carried by the copper cube to that carried by the aluminium cube. Thermal conductivity of copper is 401 W/m-K and that of aluminium is 237 W/m-K.



Solution: $R_{Al} = \frac{L}{K_{Al} A_{Al}} = \frac{3\times10^{-2}}{237\times(3\times10^{-2})^2} = 0.14\ K/W$

$R_{Cu} = \frac{L}{K_{Cu} A_{Cu}} = \frac{3\times10^{-2}}{401\times(3\times10^{-2})^2} = 0.08\ K/W$

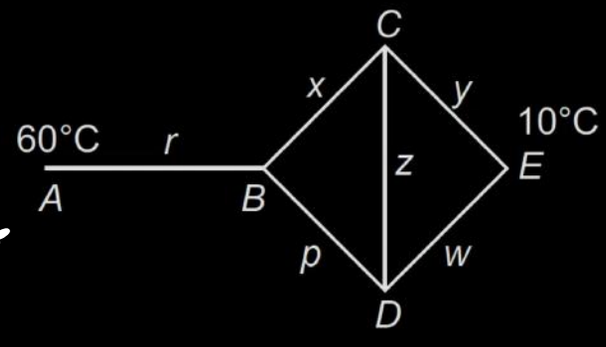
$R_{eff} = \frac{R_{Al} R_{Cu}}{R_{Al}+R_{Cu}} = 0.05\ K/W$ Now $Q = \frac{T_1 - T_2}{R_{eff}} = 1.6\times10^{3}\ W$ Ans.

(b) In parallel currents are in inverse ratio of resistances.

$\therefore\ Q_{Cu}/Q_{Al} = R_{Al}/R_{Cu} = 401/237 = 1.69$ Ans.

Node Voltage (Temperature) Method:

Example: In the adjacent figure all resistances are 1 unit except x and w which are two units each. If A and E are maintained at 60°C & 10°C, find the temperature of nodes B, C and D.



Solution: Let B, C & D represent temperatures at nodes "B", "C" and "D".
Applying junction law at these nodes:

$$\frac{B-C}{x} + \frac{B-60°C}{r} + \frac{B-D}{p} = 0 \quad (1) \quad \text{(For Node B)}$$

Similarly $$\frac{C-B}{x} + \frac{C-D}{z} + \frac{C-10}{y} = 0 \quad (2) \quad \text{(For Node C)}$$

and $$\frac{D-C}{z} + \frac{D-B}{p} + \frac{D-10}{w} = 0 \quad (3) \quad \text{(For Node D)}$$

Solving, B = 39.2°C C = 21.5°C D = 26.7°C Ans.

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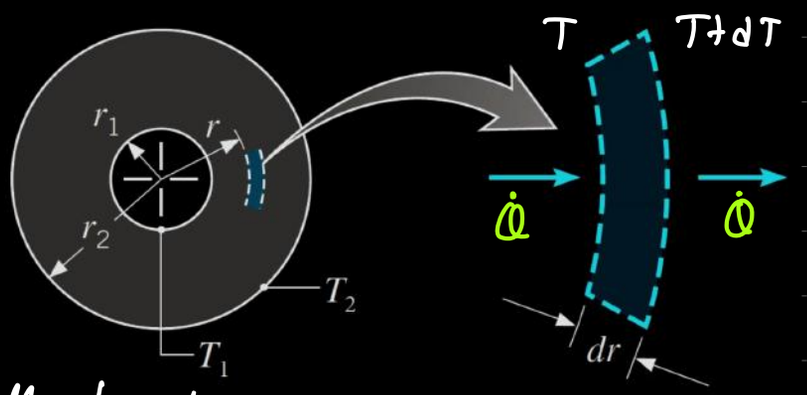
Thermal Resistance of Thick Spherical Shell:



If there is no heat generation in the sphere then in steady state, heat crossing any concentric spherical surface is same.

Now Consider a thin spherical shell element between radii r & r+dr, then:

$$\dot{Q} = -k \times 4\pi r^2 \left(\frac{dT}{dr}\right) \Rightarrow \frac{\dot{Q}}{4\pi k}\int_{r_1}^{r_2}\frac{dr}{r^2} = -\int_{T_1}^{T_2} dT \quad (1)$$

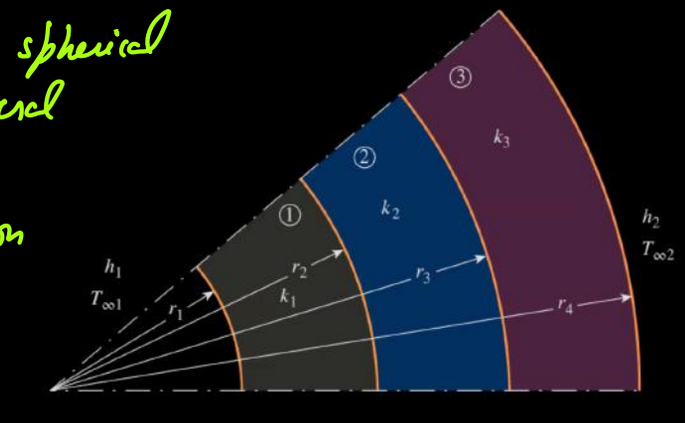
$$\Rightarrow \dot{Q} = \frac{4\pi k (T_1 - T_2)}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}, \quad \text{Now thermal Resistance} = (T_1 - T_2)/\dot{Q}$$

$$\Rightarrow \boxed{R_T = \frac{1}{4\pi k}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}$$

→ Here we assumed that k is a constant with temperature, if k were instead a function of temperature we could keep inside the temperature integral and if it were a function of r we would keep it inside the r integral.

→ If instead of a single thick spherical shell it were a combination of several concentric spherical shells, we can consider it as a series combination of all thick spherical shell resistors i.e.:

$R_{eff} = R_1 + R_2 + R_3$ etc.



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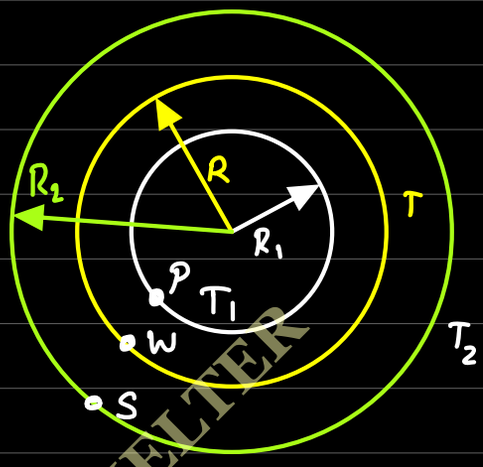
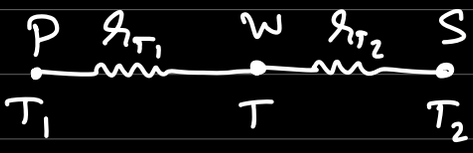
EXAMPLE (IRODOV 2.254): Find the temperature distribution in a thick spherical shell whose inner and outer radii are R_1 and R_2 and the temperatures are maintained at T_1 and T_2 respectively.

Sol: The system can be thought of a series combination of two thick shells in series i.e first between R_1 and R and the second one between R and R_2.





Now thermal resistance between P & W $= \frac{1}{4\pi K}\left(\frac{1}{R_1} - \frac{1}{R}\right) = \mathcal{R}_{T_1}$

Similarly thermal resistance between W & S $= \frac{1}{4\pi K}\left(\frac{1}{R} - \frac{1}{R_2}\right) = \mathcal{R}_{T_2}$

Now By junction law at W: $\frac{T - T_1}{\mathcal{R}_{T_1}} + \frac{T - T_2}{\mathcal{R}_{T_2}} = 0$

$$\Rightarrow T = \frac{\mathcal{R}_{T_2} T_1 + \mathcal{R}_{T_1} T_2}{\mathcal{R}_{T_1} + \mathcal{R}_{T_2}} = T_1 + \frac{T_2 - T_1}{\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}\left[\frac{1}{R_1} - \frac{1}{R}\right]$$

(Hyperbolic Variation)

→ Think (If R were at exact mid of the thickness, then will its temperature be closer to T_1 or T_2).

Sol: Since inner shell has larger resistance (due to smaller surface areas through which heat is crossing) ∴ inner shell must have greater temperature difference for same thermal current. ∴ T will be closer to T_2 at the "mid point."

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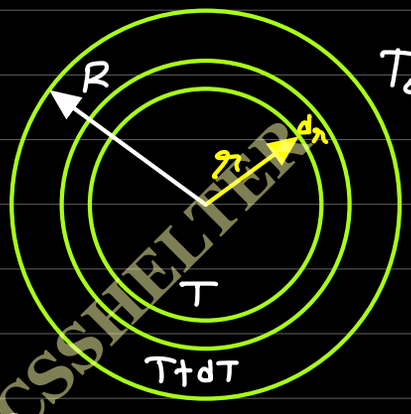
Sphere In Steady State With Heat Generation:

Example: Irodov 2.257:

The thermal power of density w is generated uniformly inside a uniform sphere of radius R and heat conductivity coefficient k. Find the temperature distribution in the sphere provided the steady-state temperature at its surface is equal to T_0.

Solution: In steady state heat generated in any volume must be equal to the heat escaping from its surface. Now consider a spherical volume of radius r with a layer of thickness dr and temperature gradient (dT/dr). Then by energy flow in steady state:

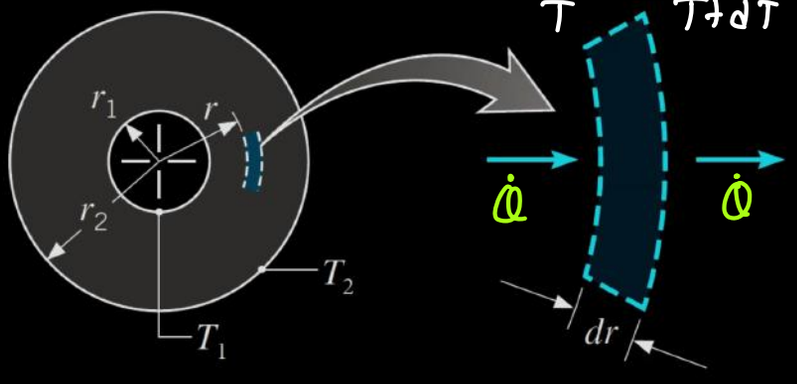


$$w \times \frac{4}{3}\pi r^3 = -k \times 4\pi r^2 (dT/dr)$$

$$\Rightarrow \frac{w}{3k}\int_R^r r\,dr = -\int_{T_0}^{T} dT \Rightarrow T = T_0 + \frac{w}{6k}(R^2 - r^2)$$ dns.

Thermal Resistance Of A Thick Cylindrical Shell (Radial Flow):

Consider a cylinder of length L carrying a radial heat flow as shown. Here also we can break the cylinder in to several concentric thin shells of thickness dr each. Each such shells carries same heat current $\therefore$ they can be considered as series connection.



Thus $$dR_T = \frac{dr}{k \times 2\pi r L} \Rightarrow R_T = \frac{1}{2\pi k L}\int_{r_1}^{r_2}\frac{dr}{r} = \frac{1}{2\pi k L}\ln\left(\frac{r_2}{r_1}\right)$$

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Now having found the thermal resistance, thermal current can be simply found as $\dot{Q} = \frac{T_1 - T_2}{R_T}$.

Cylinder With Heat Generation (Steady State Radial Flow):

Example: Irodov 2.256.

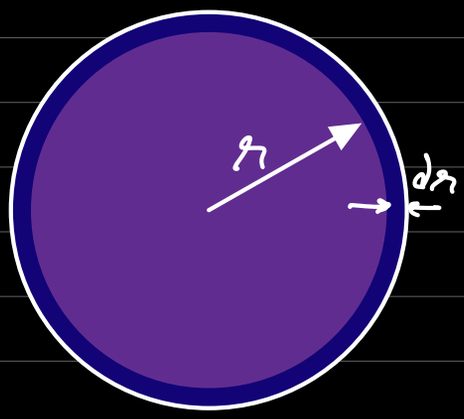
2.256. A constant electric current flows along a uniform wire with cross-sectional radius R and heat conductivity coefficient κ. A unit volume of the wire generates a thermal power w. Find the temperature distribution across the wire provided the steady-state temperature at the wire surface is equal to T_0.

Solution: Just like the case of a sphere, here also the total generation in any volume must be equal to the heat escaping from its surface. Now consider a cylindrical portion of radius r. Translating this mathematically:

$$w \times \pi r^2 L = -k \times 2\pi r L \times \frac{dT}{dr}$$

$$\Rightarrow \int_R^r \frac{w r}{2k}\, dr = \int_{T_0}^{T} -dT$$

$$\Rightarrow T = T_0 + (R^2 - r^2)\frac{w}{4\kappa} \text{ ans.}$$



Example: Irodov 2.250 (Varying Temperature Difference)

2.250. Two chunks of metal with heat capacities C_1 and C_2 are interconnected by a rod of length l and cross-sectional area S and fairly low heat conductivity k. The whole system is thermally insulated from the environment. At a moment $t = 0$ the temperature difference between the two chunks of metal equals $(\Delta T)_0$. Assuming the heat capacity of the rod to be negligible, find the temperature difference between the chunks as a function of time.

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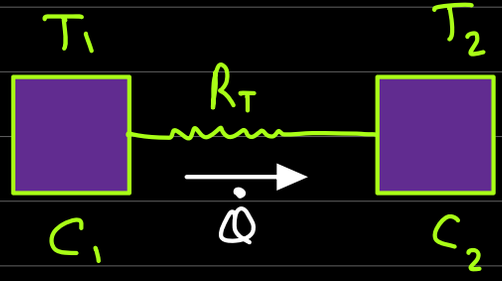
Solution: $\dot{Q} = \dfrac{(T_1 - T_2)}{R_T}$ ①



Also: $\dot{Q} = C_2 \dot{T}_2$ ② and

$\dot{Q} = -C_1 \dot{T}_1$ ③

From ② & ③: $\dfrac{\dot{Q}}{C_2} + \dfrac{\dot{Q}}{C_1} = \dot{T}_2 - \dot{T}_1 = -\dot{\Delta T}$

(Here $T_1 - T_2 = \Delta T$)

Substituting for $\dot{Q}$ from ①:

$$\frac{\Delta T}{R_T}\left[\frac{1}{C_1} + \frac{1}{C_2}\right] = -\dot{\Delta T}$$

Now this is a differential equation in ΔT and $t \Rightarrow$

$$\int_{\Delta T_0}^{\Delta T} \frac{d\,\Delta T}{\Delta T} = -\int_0^t \frac{1}{R_T}\left[\frac{1}{C_1} + \frac{1}{C_2}\right] dt \Rightarrow \Delta T = \Delta T_0\, e^{-\alpha t} \quad \text{Ans.}$$

Here $\alpha = \left(\dfrac{1}{C_1} + \dfrac{1}{C_2}\right)\dfrac{Sk}{l}$ $\left(\because R_T = \dfrac{l}{Sk}\right)$

Note that by Energy Conservation $C_1 T_1 + C_2 T_2 = C_1 T_{1_0} + C_2 T_{2_0}$ and $T_1 - T_2 = \Delta T$, using these two equations we can also find the absolute values of T_1 and T_2 if T_{1_0} and T_{2_0} were given.

EXAMPLE (IRODOV 2.251): TEMPERATURE DEPENDENT CONDUCTIVITY.

2.251. Find the temperature distribution in a substance placed between two parallel plates kept at temperatures T_1 and T_2. The plate separation is equal to l, the heat conductivity coefficient of the substance $k \propto \sqrt{T}$.

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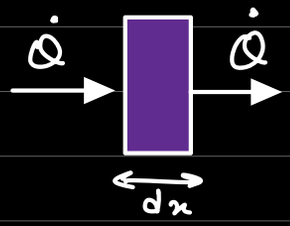
Solution: Consider an element of thickness dx having a local temperature T and thermal conductivity $K\sqrt{T/T_0}$.

Now $$\dot{Q} = -K\sqrt{\frac{T}{T_0}}\frac{dT}{dx} \quad (1)$$



Now in steady state, $\dot{Q}$ is constant. Integrating (1) appropriately

$$\int_0^x \dot{Q}\,dx = \frac{-K}{\sqrt{T_0}}\int_{T_1}^{T}\sqrt{T}\,dT \Rightarrow \dot{Q}x = \frac{-K}{\sqrt{T_0}} \times \frac{2}{3}\left[T^{3/2} - T_1^{3/2}\right] \quad (2)$$

Now @ $x = l$, $T = T_2 \Rightarrow$ $$\dot{Q} = \frac{-2K}{3\sqrt{T_0}\,l}\left[T_2^{3/2} - T_1^{3/2}\right]$$

Using this in (2):

$$\frac{-2Kx}{3\sqrt{T_0}\,l}\left[T_2^{3/2} - T_1^{3/2}\right] = \frac{-2K}{3\sqrt{T_0}}\left(T^{3/2} - T_1^{3/2}\right)$$

$$\Rightarrow T = T_1\left(1 + \frac{x}{l}\right)\left(\left(\frac{T_2}{T_1}\right)^{3/2} - 1\right)^{2/3} \quad \text{Ans.}$$

BLACK BODY RADIATION: Consider heated filament of an incandescent bulb. It emits radiation due to acceleration of charged particles. The emitted radiation contains continuous distribution of wavelengths from all portions of electromagnetic spectrum, although its exact distribution depends on the temperature of the body. [At room temperature, major fraction of this radiation is in infrared region and ∴ it does not appear to glow.]. As defined earlier, a blackbody is a surface which does not reflect any incident radiation (absorbs everything). When such a surface emitted by it is heated, the radiation emitted by it is called Blackbody Radiation.

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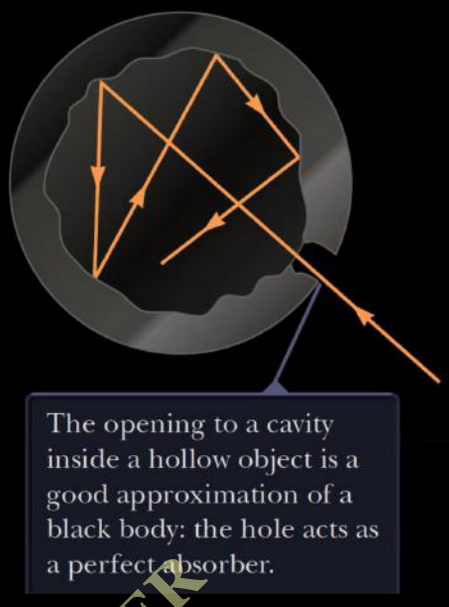
CAVITY RADIATION: A fairly good approximation of a blackbody can be created by having a small hole leading to the inside of a hollow object as shown in the figure. Any radiation incident on the hole is reflected several times on the interior walls of the cavity and thus the hole acts like a nearly perfect absorber. The radiation coming out of the hole only depends on temperature of the cavity walls and not the material of the cavity walls. The cavity radiation was studied in great detail and many scientists tried to figure out certain facts which were observed consistently in their experiments:



The opening to a cavity inside a hollow object is a good approximation of a black body: the hole acts as a perfect absorber.

① Total Power of Emitted Radiation Increases With Fourth Power of temperature of the Black Body, Mathematically:

$$P = \epsilon \sigma A T^4$$

$\epsilon = 1$ for a black body.

[Stefan, Boltzmann Law for Black Body Radiation]

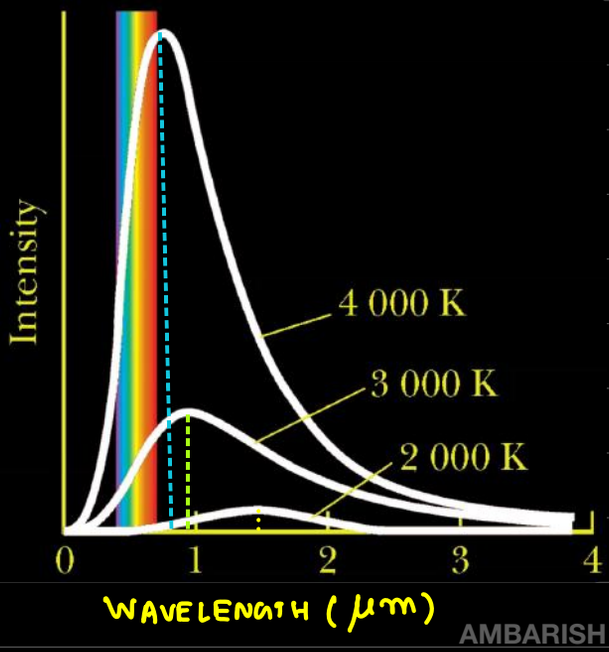
Here $\sigma = 5.67 \times 10^{-8}\ W/m^2K^4$.

$\epsilon \rightarrow$ Emissivity of the Body.

② The peak of wavelength distribution shifts to shorter wavelengths as the temperature increases. This behaviour is described by following Relationship:

$$\lambda_{max} \times T = b = 2.9\ mm\ K$$

(WIEN'S DISPLACEMENT LAW)



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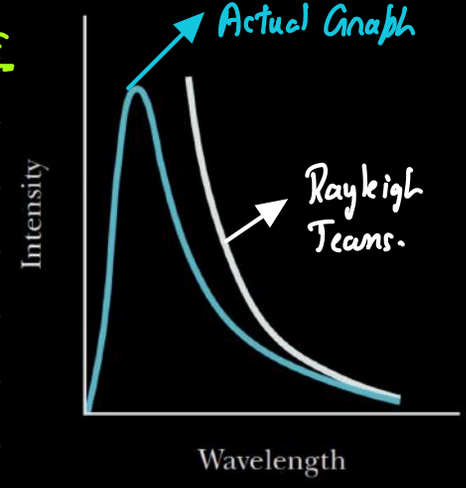
→ Near 5800 K, the graph peaks at 500 nm which is greenish yellow ∴ Sun appears yellowish. (Other colors are also present but yellow colour is present with highest intensity).

→ At Extremely high temperatures, the graph would cut nearly uniformly above all visible wavelengths and hence the word "White Heat", Hottest stars would appear white and blue stars are hotter than red stars.

→ The above graph also explains why the heater coils at first glow red before becoming yellowish, and similarly becoming gradually reddish while cooling down. (At room temperature peak occurs around infrared and ∴ no glow is observed).

Rayleigh Jeans Law And Ultraviolet Catastrophe

Spectral Emittance (Exitance) $I(\lambda)$



Power emitted per unit area per unit wavelength is known as spectral emittance ($I(\lambda)$). Thus power emitted per unit area between wavelengths λ & $\lambda + d\lambda$ is given by $I(\lambda)d\lambda$, and area under the $I(\lambda)$ vs λ graph gives total power emitted per unit area. As seen earlier in our discussion of Wien's law, $I(\lambda)$ is also a function of temperature. Based on classical ideas, Rayleigh and Jeans tried to come up with a theoretical prediction of variation of $I(\lambda, T)$ which is given as:

$$I(\lambda, T) = \frac{2\pi c k_B T}{\lambda^4}$$ ← Rayleigh Jeans Law.

This formula worked very well for high wavelength ranges (matched nicely with experimental results), however it failed miserably in the low wavelength ranges (ultraviolet).

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This observation is known as "ULTRAVIOLET CATASTROPHE" in the history of quantum mechanics. Besides considerable deviation from the graph, according to their equation, a blackbody would emit infinite energy which is obviously absurd. (The integral $I(\lambda,T)\,d\lambda$ diverges to infinity).

→ Ultraviolet catastrophe was a very important observation because it led to the beginning of quantum mechanics.

PLANCK'S RADIATION LAW: MAX Planck came up with an accurate formula describing $I(\lambda,T)$ which exactly matched with the experimental graph. He made the following assumptions to provide a theoretical basis to his formula:

1. Cavity radiation comes from atomic oscillators in cavity walls (Just like Rayleigh Jeans Model).
2. Energy of any oscillator can have only discrete values as integer multiples of $h\nu$ where ν is the frequency of oscillation and h is the planck's constant, Mathematically:

$$E_n = nh\nu.$$

Here n is a positive number known as quantum number.

3. Oscillators emit or absorb energy while making transitions from one quantum number to another quantum number state and the entire energy difference between these states is emitted as a single photon. Thus energy emitted/absorbed in transitions between adjacent energy levels is given as (say from $n=3$ to $n=2$)

$$\boxed{E = h\nu}$$

Using these ideas and some ideas from statistical mechanics somewhat similar to the ones involved in Maxwell Boltzmann distribution of gas particle velocity, Maxwell was able to derive the correct $I(\lambda,T)$ function. We shall only state the result here and omit the proof (somewhat complicated

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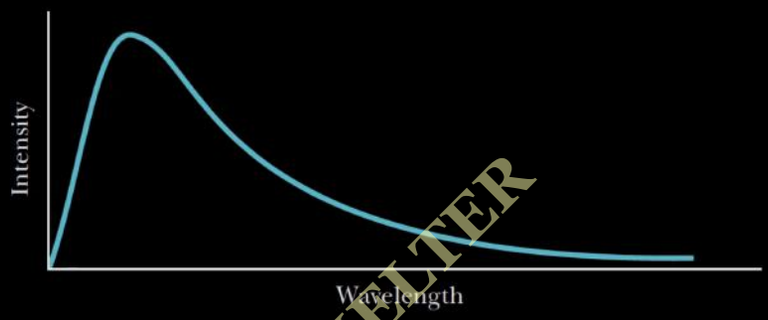
and requires some understanding of stationary waves and (combinatorics with large numbers). Planck's Radiation law is given as:

$$I(\lambda,T) = 2\pi hc^2/\lambda^5\left(e^{hc/\lambda kT} - 1\right) \quad (1)$$

One can readily see that equation (1) assumes very low values for very high wavelengths as well as very low wavelengths in agreement with the experimental graph.



→ Recall that $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \approx 1 + x$ if $x << 1$.

Using this $\left(e^{hc/\lambda kT} - 1\right) \approx hc/\lambda kT$, and using this approximation, planck's formula readily reduces to Rayleigh Jeans formula for large wavelengths.

→ Total emitted Intensity now can be found by integrating equation (1) i.e.

$$I = \int_0^\infty I(\lambda,T)\,d\lambda = \frac{2k^4\pi^5}{15c^2h^3}T^4$$

clearly this is in agreement with stefan's law i.e

$$I = \sigma T^4 \text{ with } \sigma = 2k^4\pi^5/15c^2h^3$$

thus Stefan Boltzmann law can be readily derived from Planck's radiation law.

→ One can readily find λ for which $I(\lambda,T)$ is maximum by setting the derivation of the denominator of planck's radiation law to zero. Using a CAS it yields the following result:

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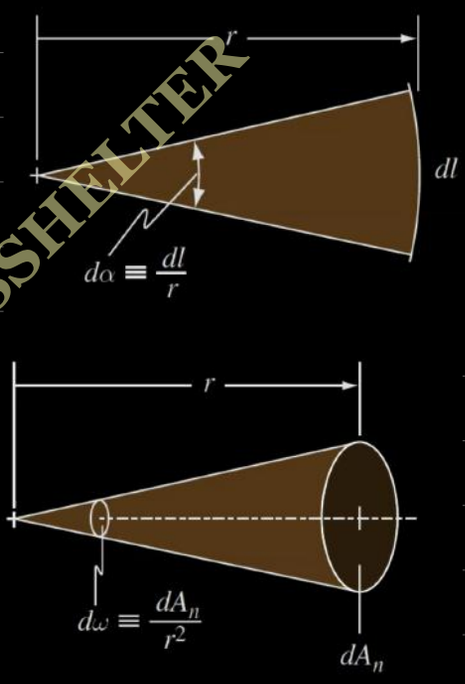
$\lambda_{max} = \frac{0.2014\ ch}{kT}$ exactly in agreement with

Wien's law i.e. $\lambda_{max} \times T = b$ (c, h and k are all constants)
Thus Stefan's law as well as Wien's law are both direct consequences of Planck's radiation law (One is obtained by Integrating it and the other one by differentiating it).

Geometry Of Black Body Radiation :

Solid Angle: Just as we define planar angle of arc of a circle at its centre as the dimensionless ratio of arc length to the radius of the circle, in the same way we define the solid angle subtended by patch on a sphere at its centre as the dimensionless ratio of area of the patch to the square of the radius. Mathematically:

$$d\omega = \frac{dA_n}{r^2} \text{ (Steradians).}$$



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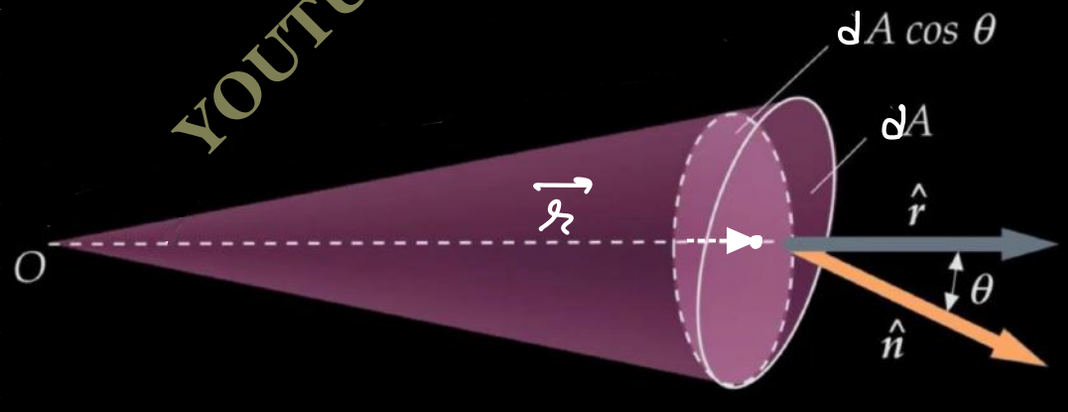
Solid Angle Subtended By Arbitrary Elemental Patch :



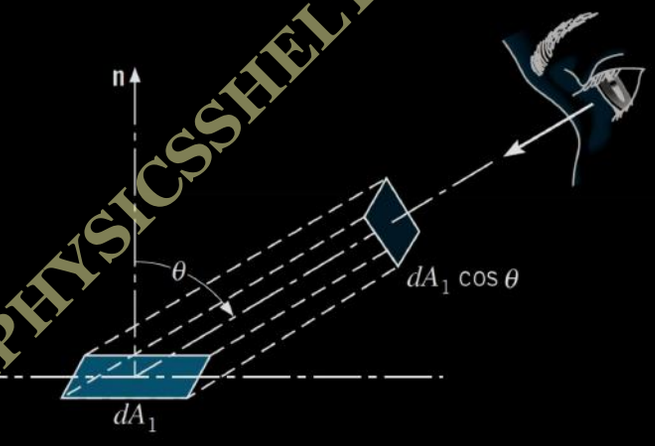
For a patch in arbitrary orientation with respect to position vector joining a point to the patch, solid angle is the ratio of patch's projected area perpendicular to its position vector, divided by square of distance from the point.

Mathematically, $d\omega = \frac{dA\cos\theta}{r^2} = \frac{\vec{dA}\cdot\hat{r}}{r^2}$

Here we have defined a vector $\vec{dA}$ as the area vector whose direction is taken as perpendicular to the patch.

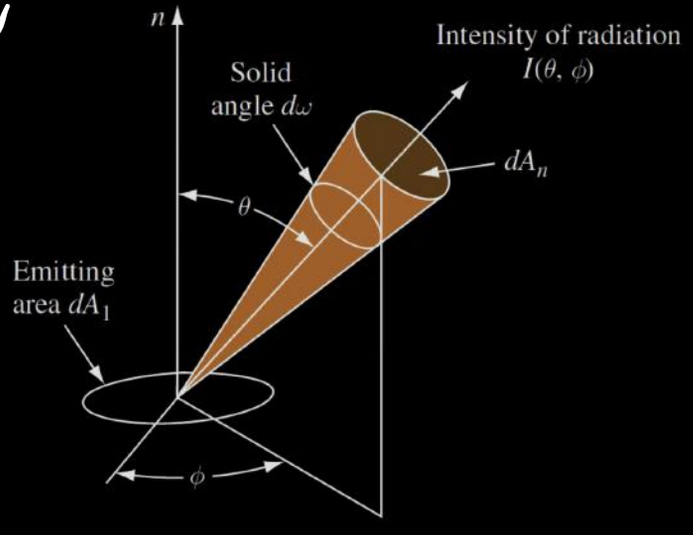
→ This can also be imagined as ratio of shadow area of the patch on a spherical screen (due to a point source of light at the centre of sphere) to the square of radius of the sphere.

AREA PROJECTED NORMAL TO RADIATION:



To understand the geometry of black body radiation we shall be shortly defining a diffuse emitter. To understand this we need to understand that apparent area of any patch depends on the direction from which we are looking. It appears maximum when we look from vertically above and reduced according to a factor of $\cos\theta$ when viewed at an angle θ from normal to the surface. i.e. $dA' = dA\cos\theta$ (see fig).

(DIRECTIONAL) INTENSITY: $(I(\theta,\phi))$



The energy emitted per unit projected area in the direction (θ,ϕ) per unit time per unit solid angle centred on that direction is called directional Intensity. Mathematically:

$$I(\theta,\phi) = \frac{dQ}{dA_1\cos\theta\, d\omega} \quad (W/m^2 Sr)$$

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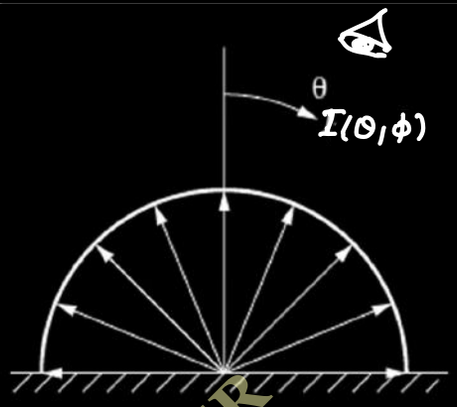
DIFFUSE EMITTER : A surface whose directional intensity is independent of the direction (θ, ϕ) is called a diffuse emitter. A diffuse emitter appears to be equally bright from every direction.

This must be understood properly. Suppose I place my eye at a distance r from the patch and view along the direction of emission, it does not mean that my eye will receive same number of photons irrespective of angle θ, rather as θ changes, projected area changes by a factor of $\cos\theta$ so also the number of photons entering the eye. Thus number of photons per unit projected area still remains the same. Mathematically it can be seen as follows:

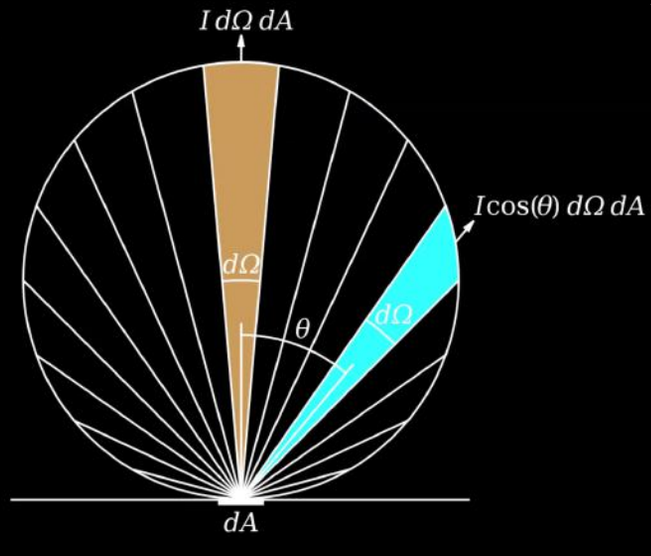


$$I(\theta,\phi) = \frac{d\dot{Q}}{dA_1 \cos\theta\, d\omega} = \text{Constant (Say } I_0\text{)}$$ then if my eye subtends a solid angle $d\omega$ on the radiating patch then power entering my eye is given as:

$d\dot{Q} = I_0\, dA_1\, d\omega \cos\theta$. Thus the power entering my eye is proportional to $\cos\theta$. (and not constant).

LAMBERT'S COSINE LAW: For a diffuse emitter if $I\, d\Omega\, dA$ is the number of photons emitted in to a vertical wedge subtending a solid angle subtending a solid angle $d\Omega$, then number of photons emitted in a wedge inclined at angle θ with vertical is given by

$$\boxed{\frac{dN}{dt} = I\, d\Omega\, dA \cos\theta}$$



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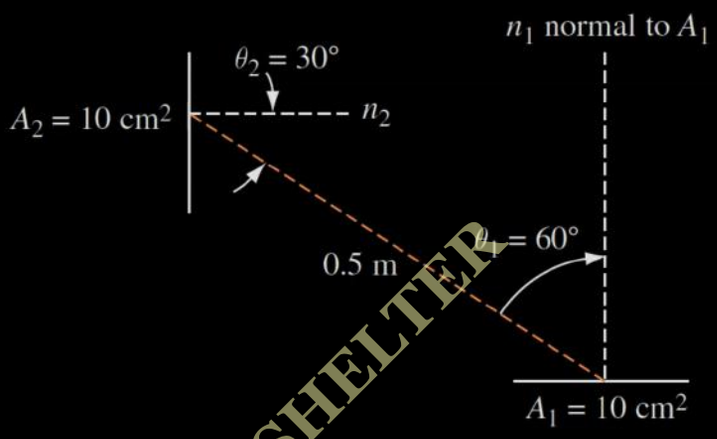
Example: (Keith & Manglik):

A flat, black surface of area $A_1 = 10\,\text{cm}^2$ emits 1000 W/m^2 sr in the normal direction. A small surface A_2 having the same area as A_1 is placed relative to A_1 as shown in Fig. 9.8, at a distance of 0.5 m. Determine the solid angle subtended by A_2 and the rate at which A_2 is irradiated by A_1.

Solution: $d\omega_{21} = \dfrac{A_2 \cos\theta_2}{r^2}$

$\Rightarrow d\omega_{21} = \dfrac{10\,\text{cm}^2 \times \cos 30^\circ}{(0.5\,\text{m})^2}$

$\Rightarrow d\omega_{21} = 0.00346$ Sr.



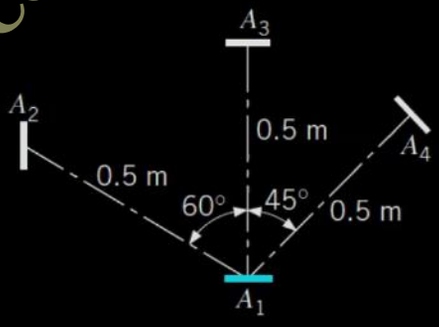
Now $d\dot{Q} = I A_1 d\omega_{21} \cos\theta_1$

$\Rightarrow d\dot{Q} = 1000 \times 10^{-3}\,\text{m}^2 \times 0.00346 \times \cos 60^\circ = 0.00173$ W Ans.

Practice Yourself (Incropera)

Example 12.1

A small surface of area $A_1 = 10^{-3}$ m^2 is known to emit diffusely, and from measurements the total intensity associated with emission in the normal direction is $I_n = 7000$ W/m$^2 \cdot$ sr.



Radiation emitted from the surface is intercepted by three other surfaces of area $A_2 = A_3 = A_4 = 10^{-3}$ m^2, which are 0.5 m from A_1 and are oriented as shown. What is the intensity associated with emission in each of the three directions? What are the solid angles subtended by the three surfaces when viewed from A_1? What is the rate at which radiation emitted by A_1 is intercepted by the three surfaces?

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Ans: (a) $I : I_n = 7000\ W/m^2/Sr$ in each direction.

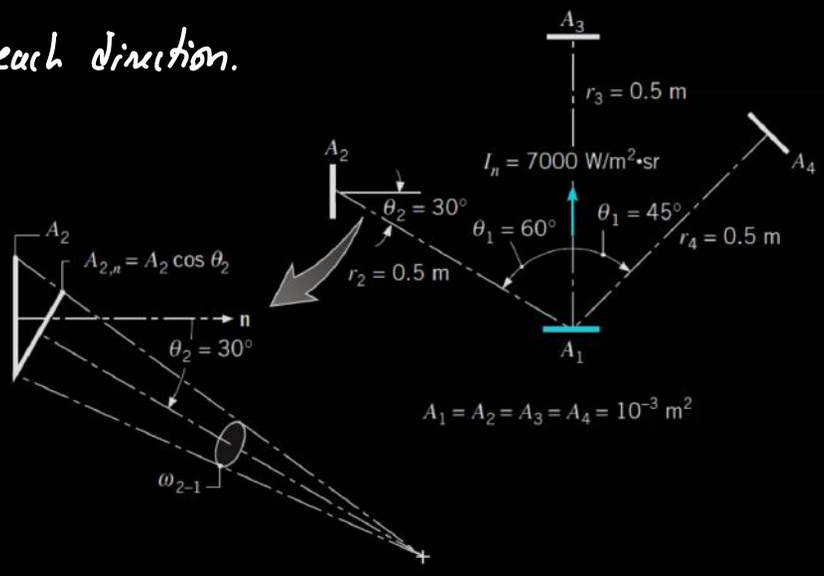
(b) $\omega_{31} = \omega_{41} = 4 \times 10^{-3}\ Sr$

$\omega_{21} = 3.46 \times 10^{-3}\ Sr.$

(c) $\dot{Q}_{1\to 2} = 12.1\ mW,$

$\dot{Q}_{1\to 3} = 28\ mW$

$\dot{Q}_{1\to 4} = 19.8\ mW$ Ans.



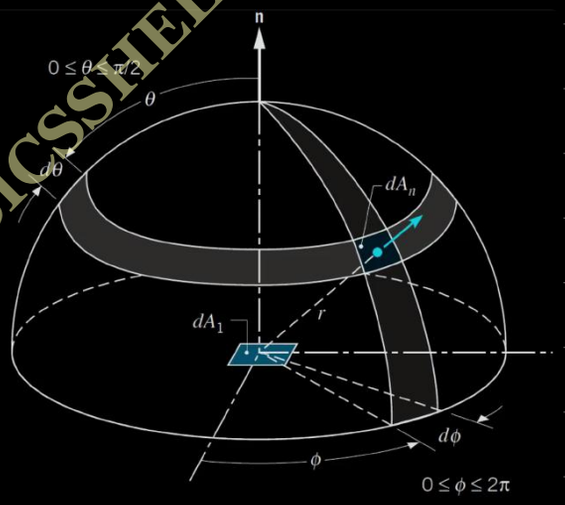
(Hemispherical) Emissive Power : Total Power Emitted by a patch in all directions is called hemispherical emissive Power.

This can be found by integrating the result of Lambert's cosine law over entire hemisphere.

Now $d\dot{Q}$ for patch

$$d\dot{Q} = I\, dA_1\, d\omega \cos\theta$$

$$\Rightarrow d\dot{Q} = I\, dA_1 \frac{dA_n}{R^2} \cos\theta$$



Now we can also find $d\dot{Q}$ for annular region Between θ & $\theta + d\theta$, $dA_n \to 2\pi R \sin\theta \times R\, d\theta$

$$\Rightarrow \int d\dot{Q} = \frac{I\, dA_1}{R^2} \cdot 2\pi R^2 \int_0^{\pi/2} \sin\theta \cos\theta\, d\theta = \pi I\, dA_1$$

We can also talk about Emittance ($\dot{Q}$ per unit area).

$$E = \pi I. \quad \Rightarrow \quad I = \frac{\sigma T^4}{\pi}, \quad I = \frac{\epsilon \sigma T^4}{\pi}$$

Grey Body: A body whose emission profile is similar to that of a black body however scaled down by some constant factor (across all wavelengths) is called a Grey Body. For grey bodies:

$\dot{Q} = \epsilon \sigma A T^4 = \epsilon A E_b(T)$ where ϵ is called emissivity of the surface, here $E_b(T)$ is emission from a blackbody at same temperature, per unit area.

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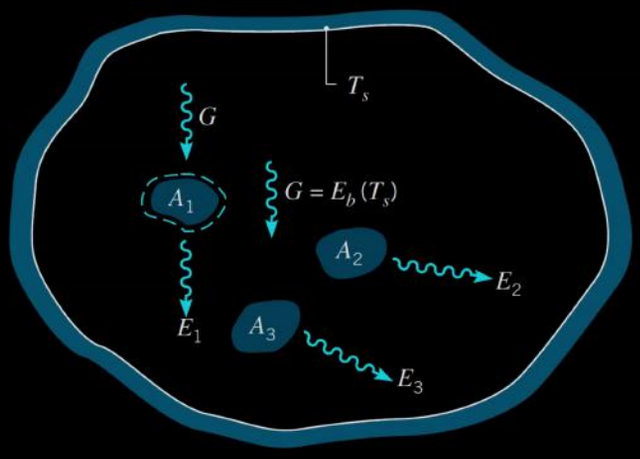
KIRCHOFF'S LAW FOR RADIATION:

Consider several Grey Bodies in a furnace with varying emissivities in thermal equilibrium with furnace temperature (Furnace can be treated like a black body (Diffuse emission)).



Now let α be the absorptivity of a patch having an area A and let G be the irradiance (falling radiation per unit area) then for thermal equilibrium:

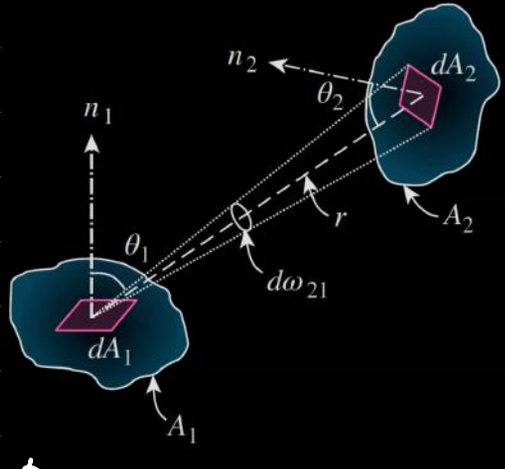
$$\alpha G A - \epsilon A E_b = 0 \quad (1)$$

Also for a Blackbody, $\alpha = \epsilon = 1 \Rightarrow GA - E_b A = 0$ (2)
(∵ considering thermal equilibrium of a black patch).

From (1), $\frac{\alpha}{\epsilon} = \frac{E_b}{G}$ from (2) $\frac{E_b}{G} = 1$

$\Rightarrow \frac{\alpha}{\epsilon} = 1$ for all Grey bodies (Thus good absorbers are good emitters and vice versa). Note that absorptivity is just a surface property and so also the emissivity.
Using a more rigorous analysis it can be shown that $\alpha = \epsilon$ is not only true considering overall powers emitted by also for powers emitted in each wavelength and in each direction).

VIEW FACTOR: Fraction of total emission emitted by a surface that is intercepted by another surface is called view factor. Mathematically f_{12} represents fraction of radiation emitted by surface 1 that is intercepted by surface 2. Obviously when sum of all the view factors of a surface is taken with all the surfaces intercepting its radiation then that sum must be 1. i.e. $\sum_{i=1}^{n} f_{1i} = 1$



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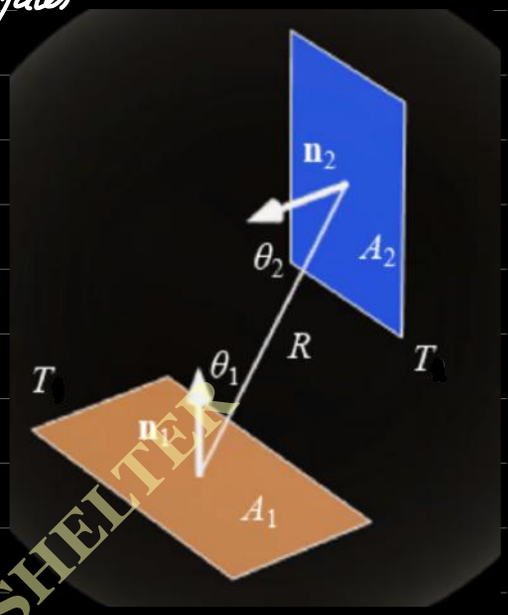
where n represents total number of surfaces with which surface is exchanging heat.

View Factor Reciprocity: For any pair of surfaces it can be shown that:

$$A_1 f_{12} = A_2 f_{21}$$



A rigorous proof of this theorem can be done by integrating the expression following Lambert's cosine rule. However we can also build an intuitive proof using concept of thermal Equilibrium. Suppose both surfaces are in thermal equilibrium with each other then they should not be exchanging any net heat with each other.

Thus $\sigma A_1 T_1^4 \times f_{12} = \sigma \times A_2 \times T_2^4 \times f_{21}$ $(T_1 = T_2)$

$\Rightarrow$ $A_1 f_{12} = A_2 f_{21}$ QED.

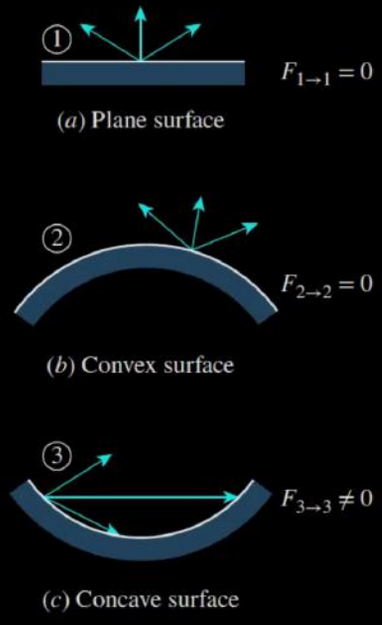
Note that view factors are purely geometry dependent as can be clearly understood from Lambert's cosine law therefore it does not matter even if we find view factor using thermodynamic arguments. (The view factor should come out to be same even if derived using Lambert's cosine law).

→ (View factor is also called shape factor by some authors).

→ Convex and flat surfaces have zero shape factors with themselves because no radiation emitted by them can fall on themselves

→ Concave surfaces have non zero shape factors with themselves as some radiation emitted by them will fall on themselves.

(This is illustrated in adjacent figure).



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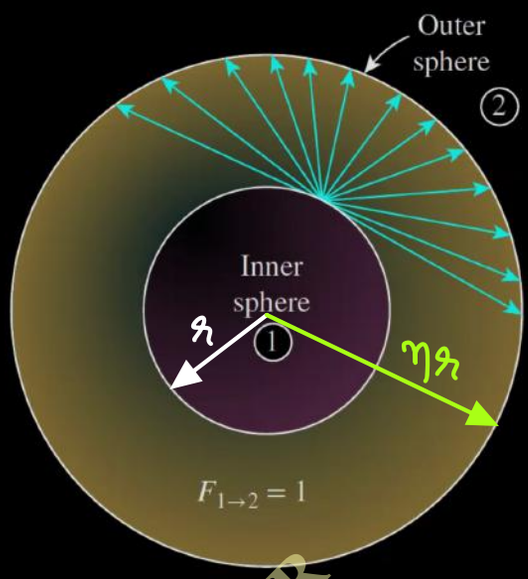
EXAMPLE: For the shown system of concentric spheres find f_{11}, f_{12}, f_{21} and f_{22}.



Solution: Since ① is convex $\therefore f_{11} = 0$ Ans.

Also, $f_{11} + f_{12} = 1$ ($\because$ completely surrounded)

$\Rightarrow f_{12} = 1$ Ans.

Now $f_{12} \times 4\pi r^2 = f_{21} \times 4\pi (\eta r)^2$

$\Rightarrow f_{21} = \frac{1}{\eta^2} \times f_{12} = \frac{1}{\eta^2}$ Ans.

Now $f_{21} + f_{22} = 1 \Rightarrow f_{22} = 1 - f_{21} = 1 - \frac{1}{\eta^2}$.

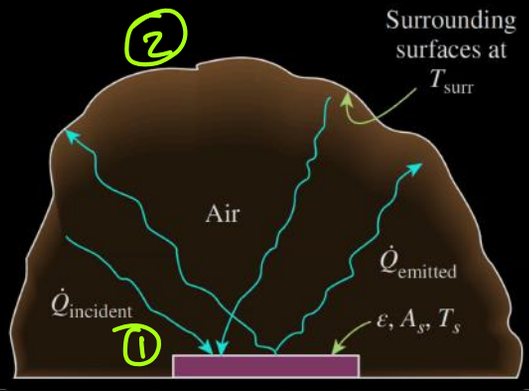
(Note that as $\eta \to 1$, $f_{22} \approx 0$).

NET HEAT EXCHANGED BETWEEN A CONVEX/PLANE SURFACE WITH BLACK ENCLOSURE.

$$\dot{Q}_{1\to 2} = \epsilon_1 \sigma A_1 f_{12} T_1^4$$

$$\dot{Q}_{21} = \alpha_1 \sigma A_2 f_{21} T_2^4$$

$$\Rightarrow \dot{Q}_{1\to 2\,net} = \dot{Q}_{1\to 2} - \dot{Q}_{21}$$



Also $\alpha_1 = \epsilon_1$ and $A_1 f_{12} = A_2 f_{21}$

$$\Rightarrow \dot{Q}_{1\to 2\,net} = \boxed{\epsilon_1 \sigma A_1 f_{12} (T_1^4 - T_2^4)}, \text{ but } f_{12} = 1$$

$$\Rightarrow \boxed{\dot{Q}_{12\,net} = \epsilon_1 \sigma A_1 (T_1^4 - T_2^4)}$$

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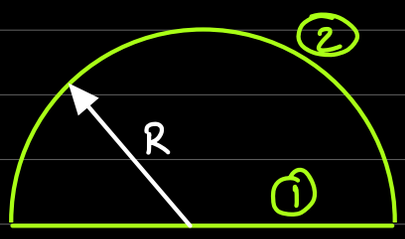
EXAMPLE: (a) Base of hemisphere is maintained at temperature T_0 while the curved surface is unpowered and surroundings may be treated as Black body at 0K. Find steady state Temperature of Curved Surface.

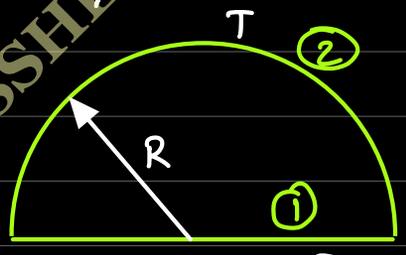


(b) Repeat the previous problem if curved surface is powered to maintain at T_0 and base is unpowered (Find Base temperature) Assume all Heat transfers through radiation alone.

Solution: (a) For any unpowered body in steady state, the power recieved through radiation must be equal to power radiated out.

$$\Rightarrow \sigma \times \pi R^2 \times (T_0^4 - T^4) = \sigma \times 2\pi R^2 (T^4 - 0^4)$$

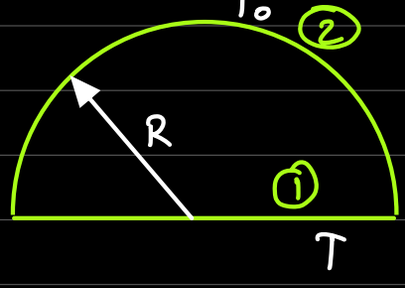
$$\Rightarrow T_0^4 = 3T^4 \Rightarrow T = T_0/3^{1/4} \text{ Ans.}$$



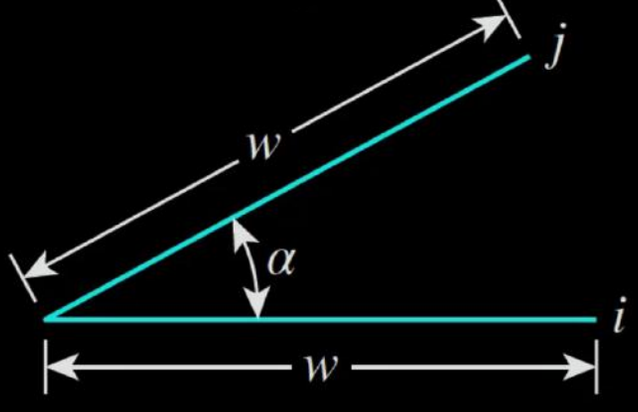
(b) For any unpowered body, net radiation going out must be zero. Thus net energy radiated from inner and outer surface of the base must sum up to zero

$$\Rightarrow \sigma \pi R^2 (T^4 - T_0^4) + \sigma \pi R^2 (T^4 - 0^4) = 0$$

$$\Rightarrow 2T^4 - T_0^4 = 0 \Rightarrow T = T_0/2^{1/4} \text{ Ans.}$$



EXAMPLE: Two very long strips each of width w are inclined at an angle α as shown in the figure prove that $f_{ij} = 1 - \sin(\alpha/2)$, Hence find the temperature of plate j if temperature of plate i is maintained T_0. Surrounding temperature is 0K.



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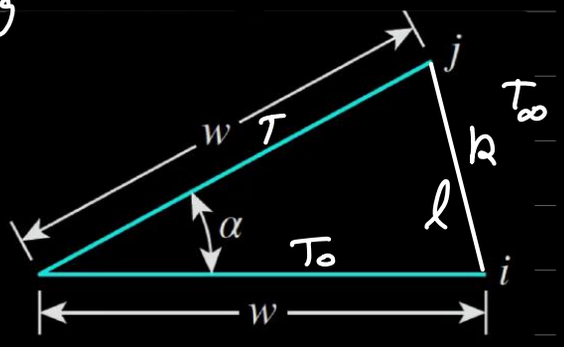
Solution: To find f_{ij} we imagine a closing surface k as shown.



Now $f_{k\to ij} = 1$ ($\because$ Complete surround)

$\Rightarrow f_{k\to i} = f_{k\to j} = \frac{1}{2}$ ($\because$ Symmetry).

Now $w \times f_{ik} = l \times f_{ki} = 2w \sin(\alpha/2) f_{ki} = w \sin(\alpha/2)$

$\Rightarrow f_{ik} = \sin(\alpha/2)$ and $f_{ik} + f_{ij} = 1 \Rightarrow f_{ij} = 1 - \sin(\alpha/2)$ QED

Now Since Plate j is unpowered, energy recieved by it must be equal to energy given out by it. Also areas are directly proportional to widths.

Thus: $\sigma w f_{ji}(T^4 - T_0^4) + \sigma w f_{jk}(T^4 - 0^4) + \sigma w (T^4 - 0^4) = 0$

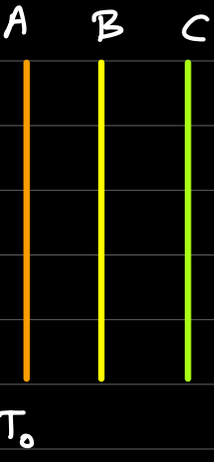
($\because$ net radiation from $j\to i = \sigma w f_{ji} \times T^4 - \sigma w f_{ij} T_0^4$)

$\Rightarrow [1 - \sin(\alpha/2)][T^4 - T_0^4] + \sin(\alpha/2) T^4 + T^4 = 0$

$\Rightarrow [1 - \sin(\alpha/2)] T_0^4 = T^4 [1 + \sin(\alpha/2) + (1 - \sin(\alpha/2))]$

$\Rightarrow T = T_0 (1 - \sin(\alpha/2))^{1/4}$ Ans.

EXAMPLE: Three very large parallel plates are arranged as shown in the figure. The left plate is maintained at T_0. Find the temperatures of plates B & C and required heating Power maintain temperature of A. (Surrounding Temperature is zero, Assume Black Bodies).



Sol: $[(T_B^4 - T_0^4) + (T_B^4 - T_C^4)]\sigma A = 0$ (1) ($\because$ B is unpowered)

Similarly: $(T_C^4 - T_B^4) + T_C^4 = 0$ (2)

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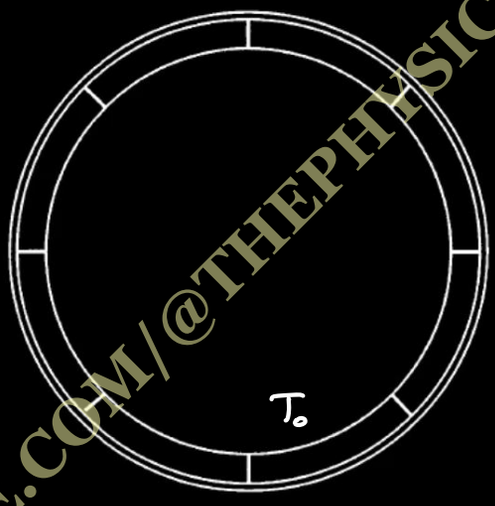
Solving ① & ②: $T_B = \left(\frac{2}{3}\right)^{1/4} T_0$, $T_C = \frac{T_0}{3^{1/4}}$ Ans.

Now $P = \sigma A\left[(T_0^4 - T_B^4) + (T_0^4 - 0^4)\right] = \sigma A\left[2T_0^4 - \frac{2}{3}T_0^4\right]$

$\Rightarrow P = \frac{4}{3}\sigma A T_0^4$ Ans.

EXAMPLE (200 PUZZLING).

P145 A totally black spherical space probe is very far from the solar system. As a result of heating by a nuclear energy source of strength I inside the probe, its surface temperature is T. The probe is now enclosed within a thin thermal protection shield, which is black on both sides and attached to the probe's surface by a few insulating rods. Find the new surface temperature of the probe. Determine also the surface temperature which would result from using N such shields.



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Solution: According to the problem: $I = \sigma A T^4$ ①

Now with one shield let temperature of probe be T_0.

$\Rightarrow I = \sigma A (T_0^4 - T_1^4)$

Also $I = \sigma A (T_1^4)$ (In steady state same heat is piercing the shield)

$\Rightarrow 2I = \sigma A T_0^4$

$\Rightarrow 2\sigma A T^4 = \sigma A T_0^4 \Rightarrow T_0 = 2^{1/4} T$

Similarly with N shields:

$I = \sigma A (T_0^4 - T_1^4)$

$I = \sigma A (T_1^4 - T_2^4)$

$\vdots$

$I = \sigma A (T_{N-1}^4 - T_N^4)$

$I = \sigma A T_N^4$

$\Rightarrow (N+1)I = \sigma A T_0^4 = (N+1)\sigma A T^4$ (1) $\Rightarrow T_0 = T(N+1)^{1/4}$ Ans.

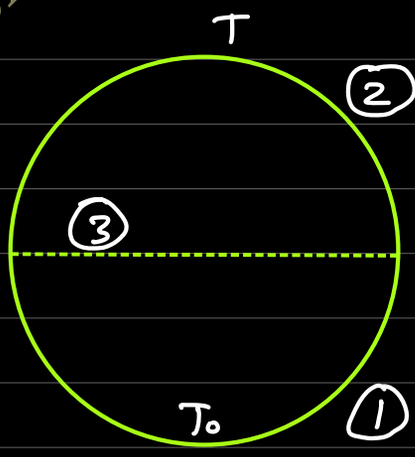
Think: In previous situation by what factor should generation be reduced if the temperature of probe it to be maintained at T still.

Solution: Let new power be $I/\eta \Rightarrow (N+1)\frac{I}{\eta} = \sigma A T_0^4 = \frac{(N+1)\sigma A I^4}{\eta}$

But $T_0 = T \Rightarrow \eta = (N+1)$ ($\because$ (1))

(Note here we have assumed area of each shield to be same, however if N is very large this is not a good approximation).

EXAMPLE: The temperature of lower hemisphere in the arrangement is maintained at T_0, find the steady state temperature of upper hemisphere assuming only radiation heat transfer and blackbodies. (Surrounding Temperature is zero).



Solution: For finding view factor, visualise an imaginary disc in equatorial plane between (1) & (2).
Every photon that reaches (2) from (1) must pass through (3) and every photon that crosses (3) coming from (1) must impinge on (2)
$\therefore f_{12}$ must be equal to f_{13}. Now by reciprocity:

$$f_{13} \times 2\pi R^2 = f_{31} \times \pi R^2 \quad \text{but } f_{31} = 1 \Rightarrow f_{13} = 1/2$$

$\therefore f_{12} = f_{13} = 1/2$. Now by energy conservation of unpowered upper hemisphere:

$$\frac{1}{2}\sigma A (T^4 - T_0^4) + \sigma A T^4 = 0 \Rightarrow \frac{3}{2}T^4 - \frac{T_0^4}{2} = 0$$

$$\Rightarrow T = T_0/3^{1/4} \text{ Ans.}$$

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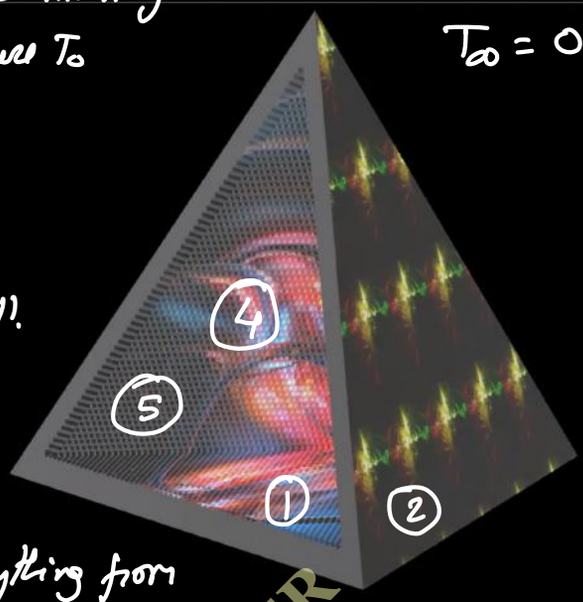
Example: One wall of a tetrahedral tent is missing while the floor is maintained at a temperature T_0. Find the temperature of the other walls. (Assume only Blackbody Radiative transfers).



Solution: By symmetry $f_{12} = 1/3 = f_{21}$

① → floor, ② wall. ④ missing Wall.
⑤ Other Wall.

$f_{24} = f_{42} = \frac{1}{3}$.

Due to symmetry, ② doesn't recieve anything from ⑤ on the net. Now Since ② is unpowered, by energy conservation:

$$\underbrace{\sigma A \times \frac{1}{3}(T_2^4 - T_0^4)}_{\text{For floor}} + \underbrace{\sigma A \times \frac{1}{3}(T_2^4 - 0^4)}_{\text{For missing Wall}} + \underbrace{\sigma A T_2^4 = 0}_{\text{Outer Surface.}}$$

$$\frac{5}{3}T_2^4 = \frac{1}{3}T_0^4 \Rightarrow T_2 = T_0/5^{1/4} \text{ Ans.}$$

Think: What would the temperature be if wall were not missing?

Sol: Considering all three walls together (Surrounding).

$$\sigma A(T_0^4 - T^4) = \sigma \times 3AT^4 \quad (\because \text{Incoming} = \text{Outgoing})$$

$\Rightarrow T = T_0/4^{1/3}$ Ans.

Applicability Of Newton's Law Of Cooling With Radiation:

Recall that we stated Newton's law of cooling as: $\dot{Q} = h(T - T_\infty)A$ where h was coefficient for convection, T temperature of the surface and T_∞ temperature of the surroundings. For small temperature differences a similar formula can be obtained for radiation heat losses also.

For radiative losses:

$$\dot{Q} = \sigma A(T^4 - T_\infty^4),$$ Now let there be a small temperature difference ΔT between T and T_∞ then we can write heat loss as:

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$$\dot{Q} = \sigma A\left((T_\infty+\Delta T)^4 - T_\infty^4\right) = \sigma A T_\infty^4\left[\left(1+\frac{\Delta T}{T_\infty}\right)^4 - 1\right]$$

Now if $\frac{\Delta T}{T_\infty} << 1$ then we can say $\dot{Q} = 4\sigma A T_\infty^3 \Delta T$

$\Rightarrow \dot{Q} = 4\sigma A T_\infty^3 [T - T_\infty]$, thus for small temperature differences we can say that even radiation losses are proportional to $(T - T_\infty)$ and we can define an h_{eff} to include all kinds of losses:

$\dot{Q} = h_{eff}(T - T_\infty)A$, here h_{eff} takes in to account conduction, convection and radiation. (One must be very cautious while applying Newton's law of cooling with radiation, due to fourth power dependence, there can be considerable errors if $\frac{\Delta T}{T_\infty}$ is not $<< 1$.

NEWTON'S LAW OF COOLING IN TERMS OF TEMPERATURE VARIATION:

Recall from Calorimetry: $\dot{Q} = mS\frac{dT}{dt}$, also $\dot{Q} = h(T - T_\infty)$

Combining the two:

$$mS\frac{dT}{dt} = -h(T - T_\infty) \text{ or } \frac{dT}{dt} = \frac{-h}{mS}(T - T_\infty)$$

Since for a given body, mS is a constant we can club all constants together and write:

$$\frac{dT}{dt} = -k(T - T_\infty) \quad (1) \quad \text{where } k \text{ is a constant}$$

depending on surface geometry, Specific heat and mass of the cooling body.

Now temperature can be found at a general time by integrating (1)

$$\int_{T_i}^{T} \frac{dT}{(T - T_\infty)} = \int_0^t -k\,dt \Rightarrow \ln\left(\frac{T - T_\infty}{T_i - T_\infty}\right) = -kt$$

$$\Rightarrow \boxed{T = T_\infty + (T_i - T_\infty)e^{-kt}} \quad (2)$$

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AVERAGE FORM OF NEWTON'S LAW OF COOLING: The solution to the differential equation ① is logarithmic as obtained in equation ②. This is generally cumbersome to do by hand in general, therefore for simpler calculations, we can use an approximate form of Newton's law of cooling given as:

$$\frac{\Delta T}{\Delta t} = -k(T_{mean} - T_\infty)$$

Here ΔT is total change in temperature over a cooling range, Δt is the time taken for change in temperature, T_{mean} is mean temperature over cooling range i.e. average of initial and final temperatures of cooling duration. This will become clearer from the following example....

EXAMPLE (HCV 28.5):

A liquid cools from 70°C to 60°C in 5 minutes. Calculate the time taken by the liquid to cool from 60°C to 50°C, if the temperature of the surrounding is constant at 30°C.

Solution: $\frac{\Delta T}{\Delta t} = -k(T_{mean} - T_\infty)$

Now for first cooling range $T_{mean} = (70° + 60°)/2 = 65°C$,
$\Delta t = 5\,min$, $\Delta T = -10°C$, $T_\infty = 30°C$ $\Rightarrow$:

$$\frac{-10}{5} = -k(65-30) \quad ①$$

Similarly for second cooling range:

$$\frac{-10}{\Delta t} = -k(55-30) \quad ② \quad \text{Now } ①/② \Rightarrow$$

$$\frac{\Delta t}{5} = (35/25) \Rightarrow \Delta t = (5 \times 35/25)\,min = 7\,min \text{ Ans.}$$

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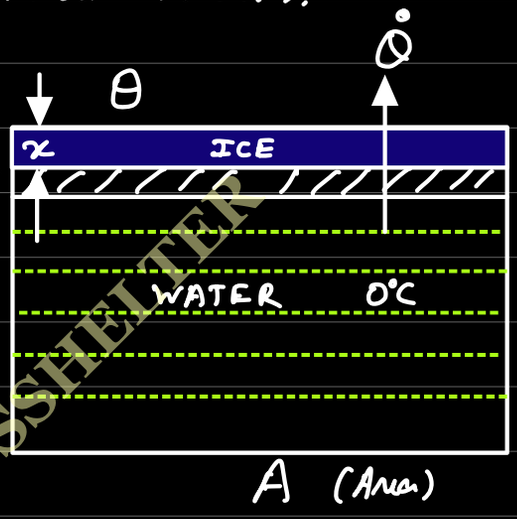
EXAMPLE: (ACCRETION OF ICE OVER A LAKE):

In extreme winters the temperature of air above a lake becomes sub zero (say θ°C). Due to this lake will begin to freeze. Let k be the thermal conductivity of ice, L be the specific latent heat of fusion and ρ be the density of ice, find how the thickness of layer of ice varies with time from the beginning of the freezing process (Assume surrounding temperature constant).



Solution: The coexistence of water and ice means a 0°C temperature just below the ice. Let the thickness of ice increase by an amount dx in a time dt then:

$$\dot{Q} = \frac{L\,dm}{dt} = L\rho A\frac{dx}{dt} = \frac{kA(0^\circ - \theta^\circ C)}{x} \quad (1)$$

Rearranging the differential equation:

$$\int_0^x \frac{L\rho}{k}\,x\,dx = -\theta\int_0^t dt$$

$$\Rightarrow \frac{L\rho}{k}\left[\frac{x^2}{2}\right] = -\theta t \Rightarrow \boxed{x = \sqrt{\frac{-2k\theta t}{L\rho}}}$$

Note that θ itself is a negative value $\therefore$ x is a Real number ☺

EXAMPLE: Find the net vector momentum emitted per unit time per unit area by a black body patch if the (directional) intensity is I hence find the ratio of emission pressure to power emitted per unit area. Use the fact that for photons $E = pc$ where E is the photon energy, p is photon momentum and c is velocity of light.

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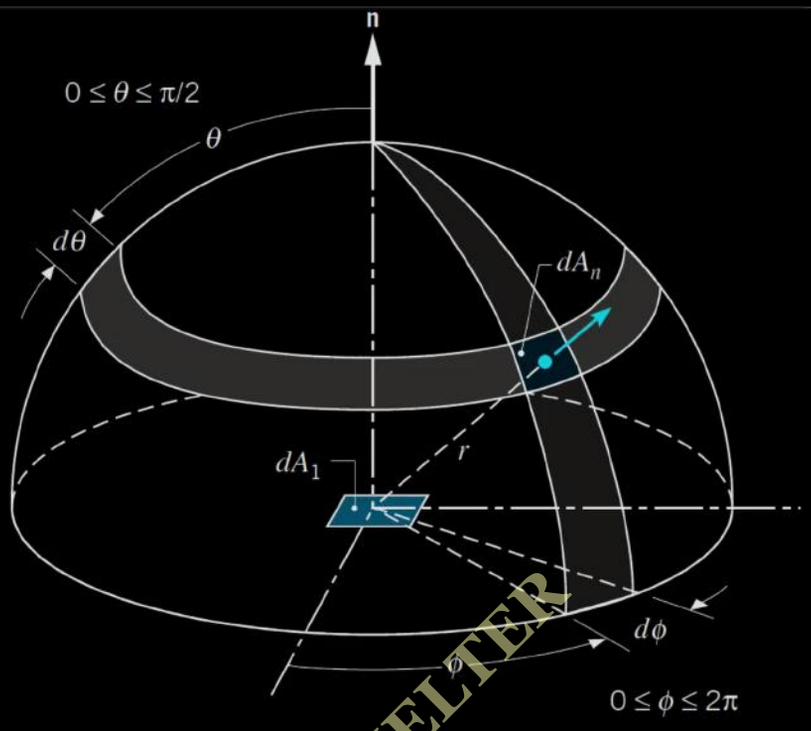
Solution: Consider an area dA_1 emitting with intensity I.

Now $d\dot{Q}$ through patch of area dA_n: $I\,dA_1\cos\theta\,d\omega$

$$\Rightarrow d\dot{Q} = I\,dA_1\cos\theta \times \frac{dA_n}{R^2}$$

Now momentum flowing through this per unit time, $d\dot{p} = \frac{d\dot{Q}}{c}$

$$\Rightarrow d\dot{p} = \frac{I\,dA_1\,dA_n\cos\theta}{R^2 c}$$



However by symmetry only vertical component will contribute to net recoil force because radial components will cancel by symmetry. Thus

$$d\dot{p}_n = \frac{I}{R^2 c}\,dA_1\,dA_n\cos^2\theta, \quad \text{for each } dA_n.$$

Now for entire cap element we will need to replace dA_n by area of the cap:

$$d\dot{p}_n = \frac{I}{R^2 c}\,dA_1\cos^2\theta \times 2\pi R\sin\theta \times R\,d\theta$$

$$\Rightarrow \dot{p}_n = \frac{2\pi I\,dA_1}{c}\int_0^{\pi/2}\cos^2\theta\sin\theta\,d\theta = \frac{2\pi}{3c}\,I\,dA_1 = F_n$$

This is the recoil force ($\because F = dp/dt$).

$$\therefore \text{Pressure} = \frac{F_n}{dA_1} = \frac{2\pi}{3c}I, \qquad \boxed{P_n = \frac{2\pi}{3}\left(\frac{I}{c}\right).}$$

Now recall Emitted power per unit area was $\frac{\dot{Q}}{A} = E_b = \pi I$.

Thus ratio of Pressure to emitted power per unit area:

$$\frac{P_n}{E_b} = \frac{2}{3c}$$

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EXAMPLE: (PATHFINDER MODERN PHYSICS, CHECK 2)

Five faces of a homogeneous cube of mass $m = 4.0$ g are painted in white (reflects all wavelengths) and one face in black (absorbs as well as radiates wavelengths) and the cube is placed in vacuum in free space at a very low temperature (near absolute zero). Initially, when the cube is at rest, it is given $Q = 900$ J of heat. As it cools down due to heat radiation, it starts moving slowly. Estimate its terminal speed. Speed of light is $c = 3 \times 10^8$ m/s.

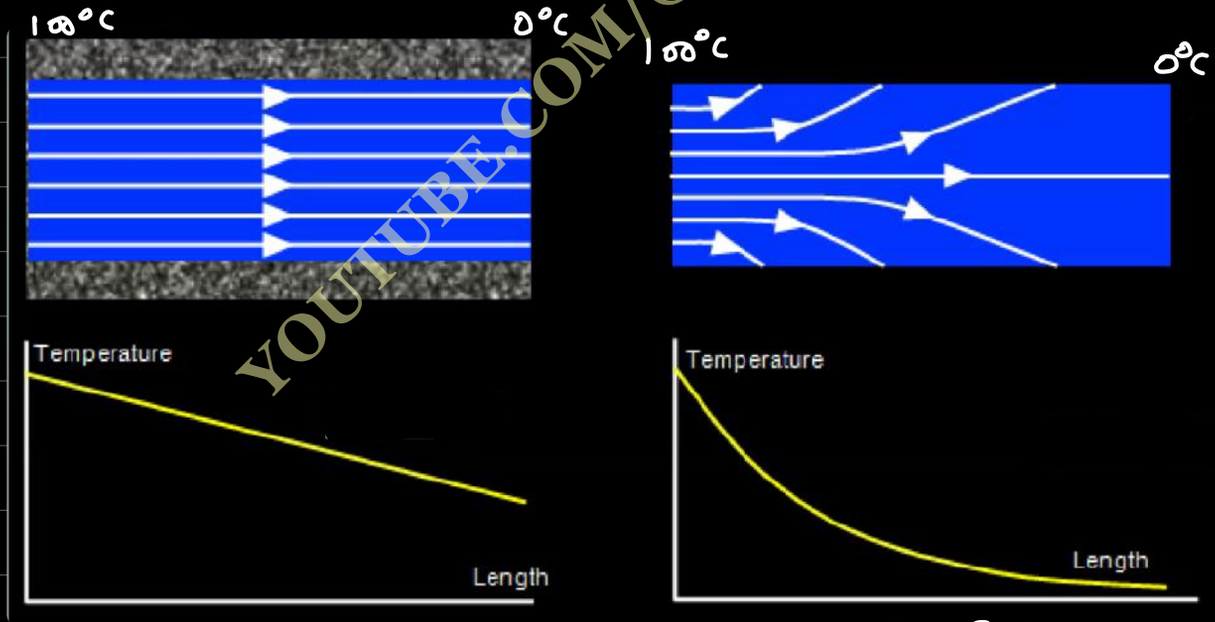
Sol: From previous example, $\frac{P_r}{E_b} = \frac{2}{3c}$ this must also be ratio of Force and Emitted power from black face and hence also the ratio of total impulse to the total emitted heat from this face

$$\therefore \quad \frac{J}{Q} = \frac{2}{3c} \Rightarrow J = \frac{2Q}{3c} = mv$$

$$\Rightarrow v = \frac{2Q}{3mc} \text{ ans.}$$

→ Answer given in pathfinder is incorrect.

TEMPERATURE GRADIENT IN LAGGED AND UNLAGGED BARS:



LAGGED BAR

UNLAGGED BAR.

In a lagged bar, there is no heat loss from lateral surface $\therefore \frac{dT}{dx}$ remains constant ($\because \dot{Q} = -kA\,dT/dx = \text{Constt}$), while in unlagged bar there is some loss from curved surface thus $\dot{Q}$ reduces

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as x increases $\therefore$ dT/dx also decreases (Assuming surroundings are at lower temperature).

Example (Pathfinder MCQ 4):

A heater of constant power is used to boil water in a pot. During heating, temperature of water increases from 60 °C to 65 °C in 1.0 min. When the heater is switched off, temperature of water falls from 65 °C to 60 °C in 9.0 min. If rate of heat dissipation to the surrounding remains almost constant, what proportion of heat received by water during heating would be dissipated to the surroundings?

(a) 9% (b) 10%
(c) 12.5% (d) 15 %

Solution: According to Newton's law of cooling $\dot{Q}_{loss} = -k(T - T_\infty)$
Let power of heater be P

$$\Rightarrow (P - \dot{Q}) \times 1\,\text{min} = mS(65° - 60°)$$

Also $$\dot{Q} \times 9\,\text{min} = mS(65° - 60°)$$

$$\Rightarrow \frac{P - \dot{Q}}{\dot{Q}} = 9 \Rightarrow 10\dot{Q} = P \Rightarrow \dot{Q} = P/10$$

$\Rightarrow$ 10% of Power is lost to surroundings. (b)

Example: Pathfinder MCQ 5:

Two identical calorimeters one containing water and the other an unknown liquid are left in a room and allowed to cool. Masses of the water, the liquid and the calorimeter are equal. Specific heat of water is 4.20 J/(g·°C) and that of the material of the calorimeters is 0.36 J/(g·°C). If the calorimeter containing water takes 60 s and that containing the unknown liquid takes 40 s in cooling from 50 °C to 45 °C, what is the specific heat in J/(g·°C) of the unknown liquid?

(a) 2.20 (b) 2.68
(c) 2.92 (d) 3.40

Solution: $$\dot{Q} = (m_w S_w + m_{cal} S_{cal})(dT/dt)_1 \quad (1)$$

Also: $$\dot{Q} = (m_l S_l + m_{cal} S_{cal})(dT/dt)_2 \quad (2)$$

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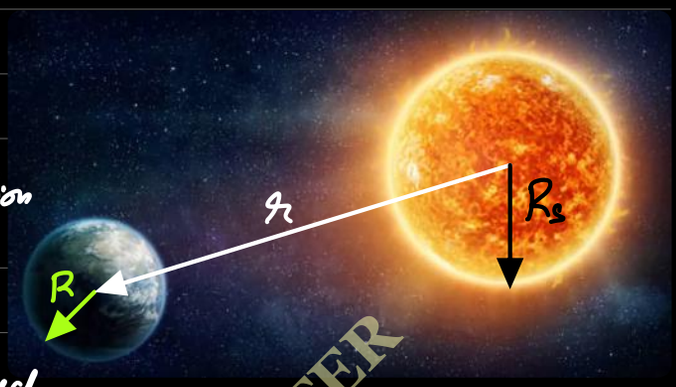
$$\frac{①}{②} \Rightarrow 1 = \frac{m(s_w + s_{cal}) \times (5/60)}{m(s_l + s_{cal}) \times (5/40)} \quad \text{Solving } s_l = \frac{2s_w - s_{cal}}{3}$$

$$\Rightarrow s_l = \frac{2 \times 4.2 - 0.36}{3} = 2.68 \text{ J/g°C} \quad \text{(b)}$$

SOLAR CONSTANT : Energy recieved by earth's surface per unit perpendicular surface to the incident solar radiation is called solar constant. Mathematically $S = 1.366 \text{ kW/m}^2$. Since sun is very far, its rays can be considered nearly parallel on earth's surface and total energy recieved by earth's surface can be estimated by multiplying solar constant by earth's projection area ⊥ to incident radiation i.e. πR^2

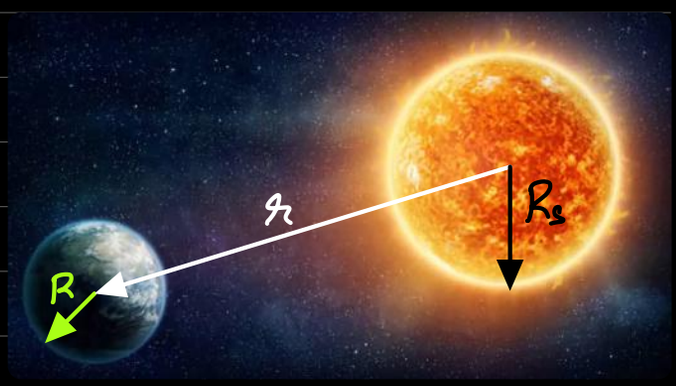


Thus incident power on earth: $P_i = S \times \pi R^2$. S itself can be estimated by considering sun as a blackbody of temperature T_s and radius R_s. The radiation per unit area recieved from sun at a distance r is given as:

$$S = \frac{\sigma \times T_s^4 \times 4\pi R_s^2}{4\pi r^2} \Rightarrow S = \sigma \left(\frac{R_s}{r}\right)^2 T_s^4.$$

EXAMPLE: Estimate the temperature of earth in terms of parameters R_s, r, R and T_s. Assume earth to be a blackbody with uniform temperature.

Sol: By energy conservation:



$$S \times \pi R^2 = \sigma \times T_e^4 \times 4\pi R^2$$

$$\Rightarrow T_e = \left(\frac{S}{4\sigma}\right)^{1/4}$$

$$\Rightarrow T_e = \left(\frac{1}{4} \times \left(\frac{R_s}{r}\right)^2 \times T_s^4\right)^{1/4}$$

$$\Rightarrow T_e = T_s \sqrt{\frac{R_s}{2r}} \quad \text{Ans.}$$

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EXAMPLE: If T_e is earth's temperature, estimate the temperature of a planet whose distance from sun is η times the distance between earth and sun. Hence estimate temperature of mercury knowing that earth is four times more distant from sun than mercury. ($T_e = 290K$)

Solution: From previous example, $T \propto 1/\sqrt{R}$ thus if a planet is η times more distant from sun than earth then its temperature is $1/\sqrt{\eta}$ times that of earth. $\Rightarrow$

$$\boxed{T_p = T_E/\sqrt{\eta}}$$

For mercury, $T_m = 290 \times \sqrt{4} = 580K$.

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